Difference of Lunar Regolith Thermal Conductivity K_d at the Apollo 15 and 17 Heat-Flow Boreholes

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Abstract

Apollo 15 and 17 are the only sites on the Moon where subsurface regolith temperatures have been measured in situ. The restored 1971–1977 Heat-Flow Experiment (HFE) record is the sole direct constraint on the meter-scale regolith thermal conductivity K_d, yet current thermal models apply a single Moon-wide value (Hayne et al., 2017; Martínez and Siegler, 2021). We retrieve K_d separately at each borehole from the restored record (Nagihara et al., 2018) by sweeping it against the meter-scale-sensor RMSE (80–234 cm) under the Hayne et al. (2017) form, excluding borestem-contaminated shallow sensors. Conditional on the published basal heat fluxes (21 and 16 mW m^{-2}; Langseth et al., 1976), the retrieval yields $K^*_{d,\mathrm{A15}} = 4.60^{+0.86}_{-0.21}$ and $K^*_{d,\mathrm{A17}} = 7.08^{+0.53}_{-0.52}$ mW m^{-1} K^{-1} (1σ bootstrap), reducing the sensor RMSE by 8% at Apollo 15 and 55% at Apollo 17 relative to the global $K_d = 3.4$. The inter-site contrast, $\Delta K^*_d = 2.3^{+0.7}_{-0.9}$ mW m^{-1} K^{-1} (95% CI [-0.1, 3.6]), marginally includes zero ($P_\mathrm{boot} \approx 0.03$); the ordering $K^*_{d,\mathrm{A17}} > K^*_{d,\mathrm{A15}}$ nonetheless survives the basal-flux reanalysis ($\geq$99% across the published Q_b envelope), though the absolute values do not. Both retrievals lie within porosity, composition, and orbital benchmarks and supply the meter-scale boundary condition needed by subsurface radiative-transfer retrievals such as the upcoming TSUKIMI terahertz mission.

Plain Language Summary. *The Moon's interior is insulated from the daily surface heating by the compacted regolith (broken rock and dust) in roughly the top meter. The numbers describing that insulation come from formulas fit to satellite measurements of the surface temperature, not to any direct sub-surface data. Sub-surface temperatures were measured at only two locations — by thermometers the Apollo 15 and 17 astronauts emplaced up to 1.4 and 2.3 m deep in 1971–1972 — and the complete records were recently recovered from the original mission tapes. We use those records to measure, at each borehole, the regolith conductivity about a meter down, where the daily temperature swing has died away and the grain-to-grain part of the conductivity has reached its deep, constant value. The Hayne (2017) global value misses these temperatures by about 1.1 K at Apollo 15 and 0.9 K at Apollo 17; refitting each site separately halves the Apollo 17 mismatch, and the two sites disagree on the conductivity by roughly a factor of 1.5. These site-specific values provide the sub-surface temperature input that upcoming lunar mapping missions need to interpret what their instruments see through the regolith.*

Key Points

- Meter-scale regolith conductivity K_d is retrieved per site from the restored Apollo 15 and 17 heat-flow records (80–234 cm).
- Apollo 15 gives $K_d = 4.60^{+0.86}_{-0.21}$ and Apollo 17 gives $7.08^{+0.53}_{-0.52}$ mW m^{-1} K^{-1} (1σ bootstrap).
- Both sites reject the Hayne et al. (2017) global K_d=3.4 at 95%; the contrast is marginal (p≈0.03) but its ordering is robust (≥99%).

1 Introduction

Lunar sub-surface thermal modeling presently relies on globally averaged conductivity parameterizations. We use the term *meter-scale regolith* throughout to mean the subsurface layer constrained by the Apollo HFE temperature sensors used in this retrieval—specifically, 80–139 cm for Apollo 15 and 130–234 cm for Apollo 17. At these depths the daily and annual thermal skins ($\lesssim 10$ cm and ~ 50 cm respectively) have damped to negligible amplitude, and the conductivity in both standard parameterizations has converged on its depth-independent asymptote K_d; under the Hayne (2017) form, the same K_d value extrapolates to all greater depths within the regolith column. The meter-scale regolith conductivity is supplied by formulations calibrated against orbital Diviner brightness temperatures: the smooth-exponential $K(T, z)$ of Hayne et al. (2017), in which a single meter-scale asymptote K_d is fit to a global average and applied moon-wide, and the temperature- and density-dependent $K(T, \rho)$ of Martínez and Siegler (2021), which extends the same functional family with additional physical content but is itself calibrated globally and applied uniformly. Both parameterizations probe only the diurnal skin. The extent to which the resulting global K_d represents any individual landing site, and the magnitude of any site-to-site departure, cannot be determined from orbital data alone.

Direct in-situ data exist only from the Apollo Heat-Flow Experiment (HFE). At Apollo 15 and 17, fiberglass borestems emplaced by the astronauts carried thermometers to depths of up to 1.4 m (Apollo 15) and 2.3 m (Apollo 17) (Langseth et al., 1972, 1976); these two boreholes remain the only lunar locations at which a sub-surface temperature profile has been measured (Fig. 1). The restored 1971–1977 HFE record (Nagihara et al., 2018) provides the data used here. Each borehole also carries a published basal heat flux (Langseth et al., 1976; Nagihara et al., 2018) which, together with the local K_d, sets the meter-scale temperature gradient. The site-specific inputs (latitudes, albedos, emissivities, basal fluxes) are collected with their original sources in Table 1.

We retrieve K_d separately at each site by holding the Hayne et al. (2017) functional form fixed and sweeping K_d alone against the meter-scale-sensor RMSE. The retrieval is placed in context by two additional forward evaluations against the same meter-scale-sensor set: the Hayne et al. (2017) form at its published global K_d, and the Martínez and Siegler (2021) $K(T, \rho)$ model at its published coefficients (Table 2). Neither published model is refit here, so the Martínez and Siegler (2021) comparison quantifies the performance of the published global $K(T, \rho)$ at each Apollo site without per-site freedom. A full per-site retrieval under the Martínez and Siegler (2021) form requires a different choice of free parameter from K_d and is deferred to a follow-on study (§4.3).

The per-site K_d^* values matter for any sub-surface radiative-transfer (RTM) retrieval that observes the Moon at wavelengths penetrating below the diurnal skin: future microwave and terahertz sub-surface sounders all need a meter-scale temperature column $T(z)$ as a forward-model boundary condition that orbital instruments cannot measure directly. One concrete example is the upcoming TSUKIMI mission (Lunar Terahertz Surveyor for Kilometer-scale Mapping, Kasai et al., 2022), whose forward model requires the sub-diurnal-skin temperature column that only the in-situ K_d can anchor.

Below the diurnal skin the modeled column is controlled by K_d through both the rectified lunation mean and the steady gradient Q_b/K_d, so the $T(z)$ ingested by any such retrieval inherits its K_d dependence; a factor-of-two error in K_d propagates through the meter-scale temperature profile and into any retrieval that uses it. The values reported here, evaluated in the Hayne et al. (2017) thermal column, supply that boundary condition from the only in-situ data available.

The retrieval is driven by an idealized synodic insolation cycle evaluated at each site's latitude (§2); because only stability-window mean temperatures enter the fit, sub-cycle timing plays no role, and the neglected ephemeris terms (solar declination, orbital eccentricity) perturb the cycle-mean insolation at or below the percent level (§2). The remaining sections present the data and methods (§2), the per-site retrieval with the headline RMSE table (§3), the interpretation with its uncertainty limits (§4), and the conclusions.

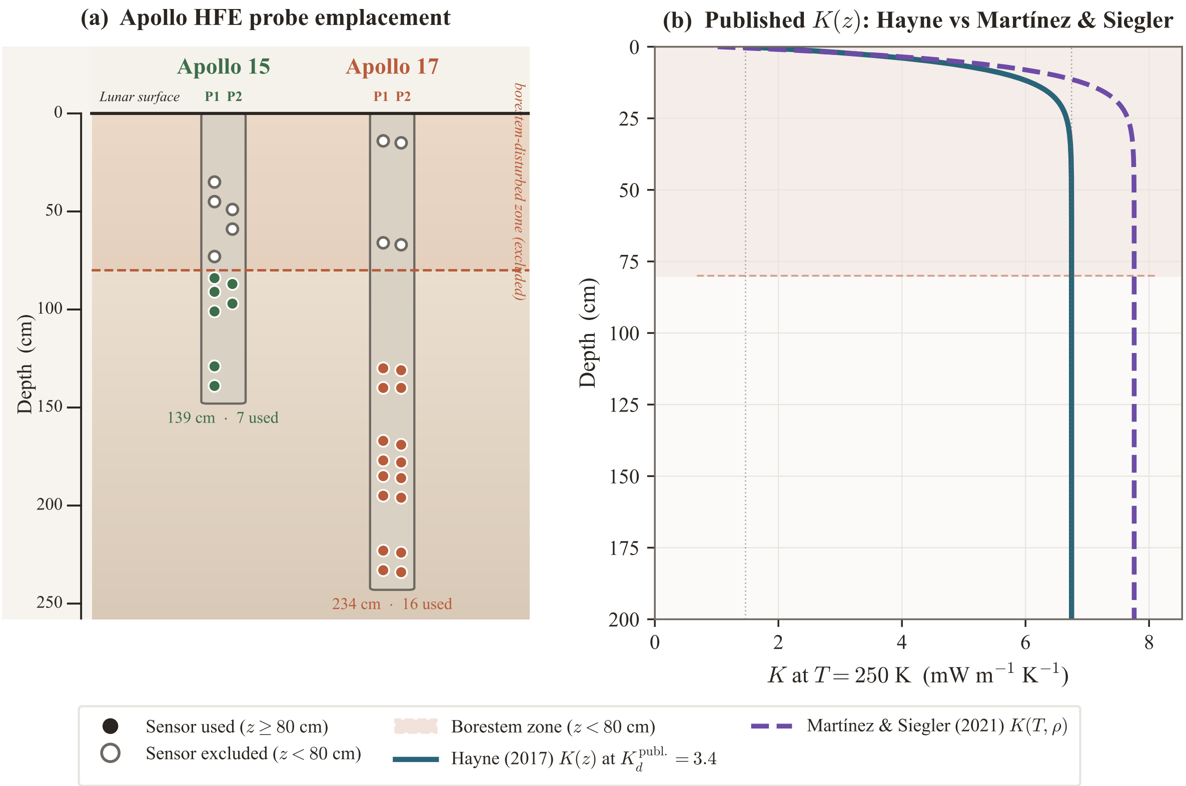

Figure 1. Apollo HFE measurement geometry and the two published $K(z)$ forms tested. **(a)** Borestem emplacements at both sites, to a shared depth scale: markers are the real sensor depths, filled where a sensor enters the retrieval ($z \geq 80$ cm) and open in the borestem-disturbed zone ($z < 80$ cm, coral) that is excluded; the two probe strings per site (P1 left, P2 right) are offset. Apollo 17 reaches 2.3 m (16 retained sensors), Apollo 15 1.4 m (7). **(b)** Conductivity profiles $K(z)$ at $T = 250$ K: Hayne et al. (2017) at $K_d^{\text{publ.}} = 3.4$ mW m^{-1} K^{-1} (solid teal) and Martínez and Siegler (2021) $K(T, \rho)$ (dashed violet), sharing the same Hayne-form $\rho(z)$.

2 Data and Methods

2.1 Apollo HFE meter-scale-sensor temperatures

The retrieval requires one equilibrium temperature T_{eq} per sensor. The restored 1971–1977 record (Nagihara et al., 2018) provides this quantity once the early portion of each record, which contains residual transients from drilling and emplacement, documented mission-operations disturbances (Nagihara et al., 2018), and slow instrumental drift, is excluded. For each sensor we extract automatically the longest trailing segment that is statistically indistinguishable from a flat steady state and average within it.

For each sensor, candidate start points covering 55–80% of the record (13 equally-spaced candidates; those retaining less than 20% of the record are excluded by a minimum-tail guard, capping the effective range at 80%) are evaluated by ordinary-least-squares linear fit to all temperatures between the candidate point and the end of the record. The accepted *stability window* is the earliest candidate for which the linear-fit slope satisfies $|d\langle T\rangle/dt| < 0.08\,\text{K/yr}$ ($\approx 2.2 \times 10^{-4}\,\text{K/d}$): the longest tail consistent with thermal equilibrium at this stringency. The adopted threshold is approximately 4.6× stricter than the $10^{-3}\,\text{K/d}$ value sometimes used in HFE analyses, and was chosen to suppress bias from residual post-disturbance drift. If no candidate satisfies the criterion, the last 25% of the record is used as a fallback. The strict criterion is met by 3 of 7 retained sensors at A15 and 6 of 16 at A17; the remainder take the fallback window, with residual trailing slopes of 0.11–0.47 K yr^{-1}. The threshold sweep below bounds the sensitivity to the choice of criterion, but residual common-epoch drift itself is not propagated as a systematic, and the within-window scatter reported as the per-sensor uncertainty characterizes short-term stability only — it does not weight the headline RMSE or the bootstrap (§4.5). We report the window-mean equilibrium

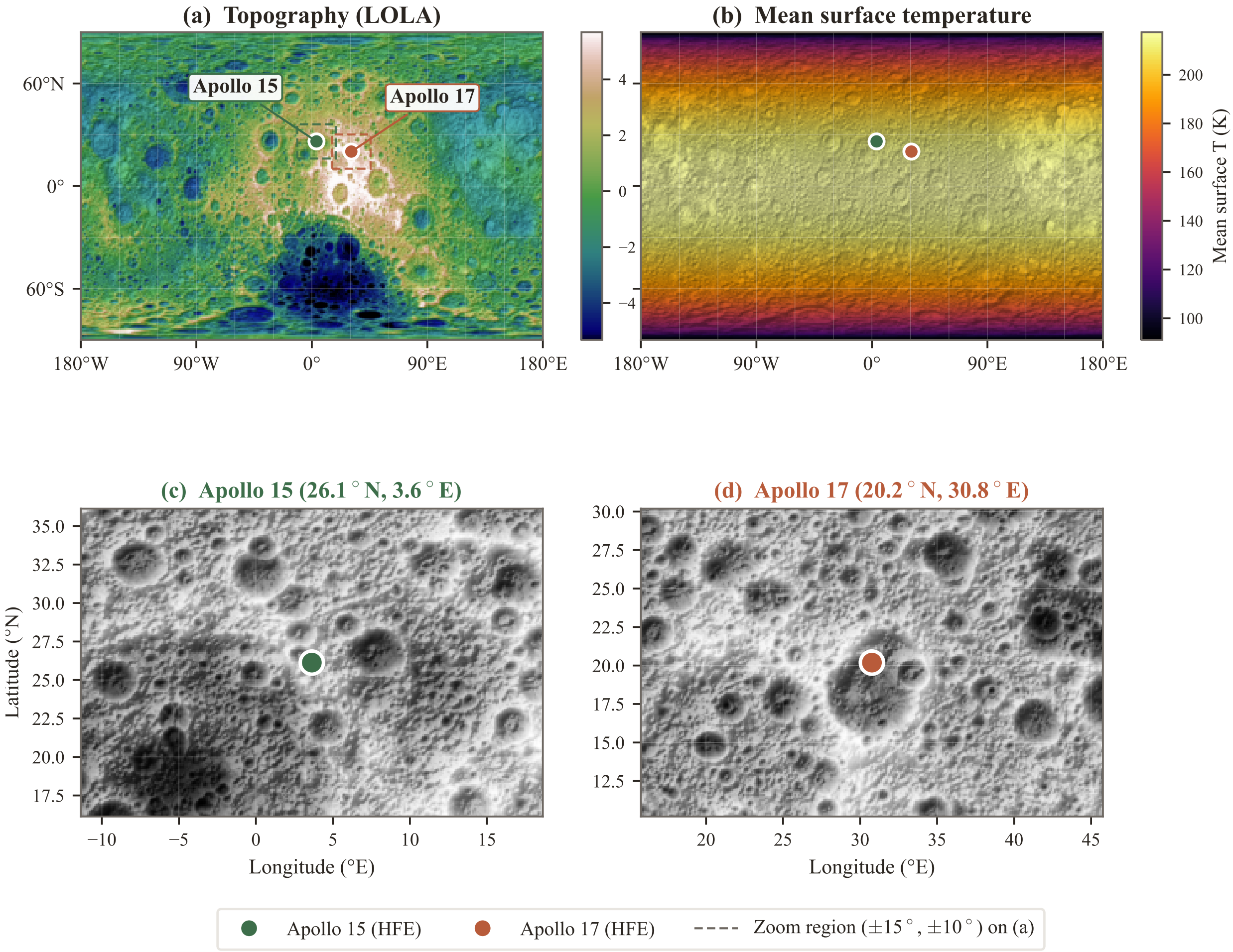

Figure 2. Apollo Heat-Flow Experiment landing-site context. **(a)** Global LOLA topography and **(b)** modeled mean surface temperature, each with the two boreholes marked; dashed rectangles on (a) show the zoom regions. **(c,d)** LOLA shaded-relief zooms (±15° longitude, ±10° latitude) resolving the Imbrium–Apennine mare/highland margin at Apollo 15 and the Taurus–Littrow highlands at Apollo 17 (Korotev, 2005; Wieczorek et al., 2006; Crawford, 2015). Borehole coordinates from the ALSEP gazetteer.

temperature $T_{\mathrm{eq}} = \langle T(z)\rangle_{\Delta t}$ together with its within-window standard deviation, which serves as the per-sensor measurement uncertainty in downstream statistics. Sensors with central depth $z < 80\,\mathrm{cm}$ (the borestem-contaminated zone; §2.4) are plotted but excluded from the RMSE.

This construction should not be read as a claim that every archived Apollo trace had reached exact zero-drift steady state. It is an estimator for the undisturbed periodic-equilibrium temperature of the regolith after the borehole disturbance has vanished. In the nominal reduction the strict flat-trend criterion is satisfied by 3/7 Apollo 15 and 6/16 Apollo 17 meter-scale sensors; the remaining retained sensors use the late-record fallback tail and therefore carry residual post-emplacement drift as a systematic component of T_{eq}. The forward model below is thus compared to equilibrium-temperature estimates, not to raw instantaneous measurements.

Figure 3 shows the full per-probe temperature record at each site, with the documented disturbance intervals shaded (Nagihara et al., 2018), the selected stability windows marked, and the fitted long-term drift $d\langle T\rangle/dt$ reported per sensor.

The retrieval is insensitive to the precise threshold. Sweeping the threshold over 0.04–0.16 K/yr (a factor of four around the adopted value) shifts K_d^* by at most $0.02\,\mathrm{mW/(m\,K)}$ at A15 (0.4% of nominal) and $0.05\,\mathrm{mW/(m\,K)}$ at A17 (0.6%), in both cases an order of magnitude below the bootstrap 1σ uncertainty (§3.2). The adopted threshold therefore lies within the flat band of the threshold response.

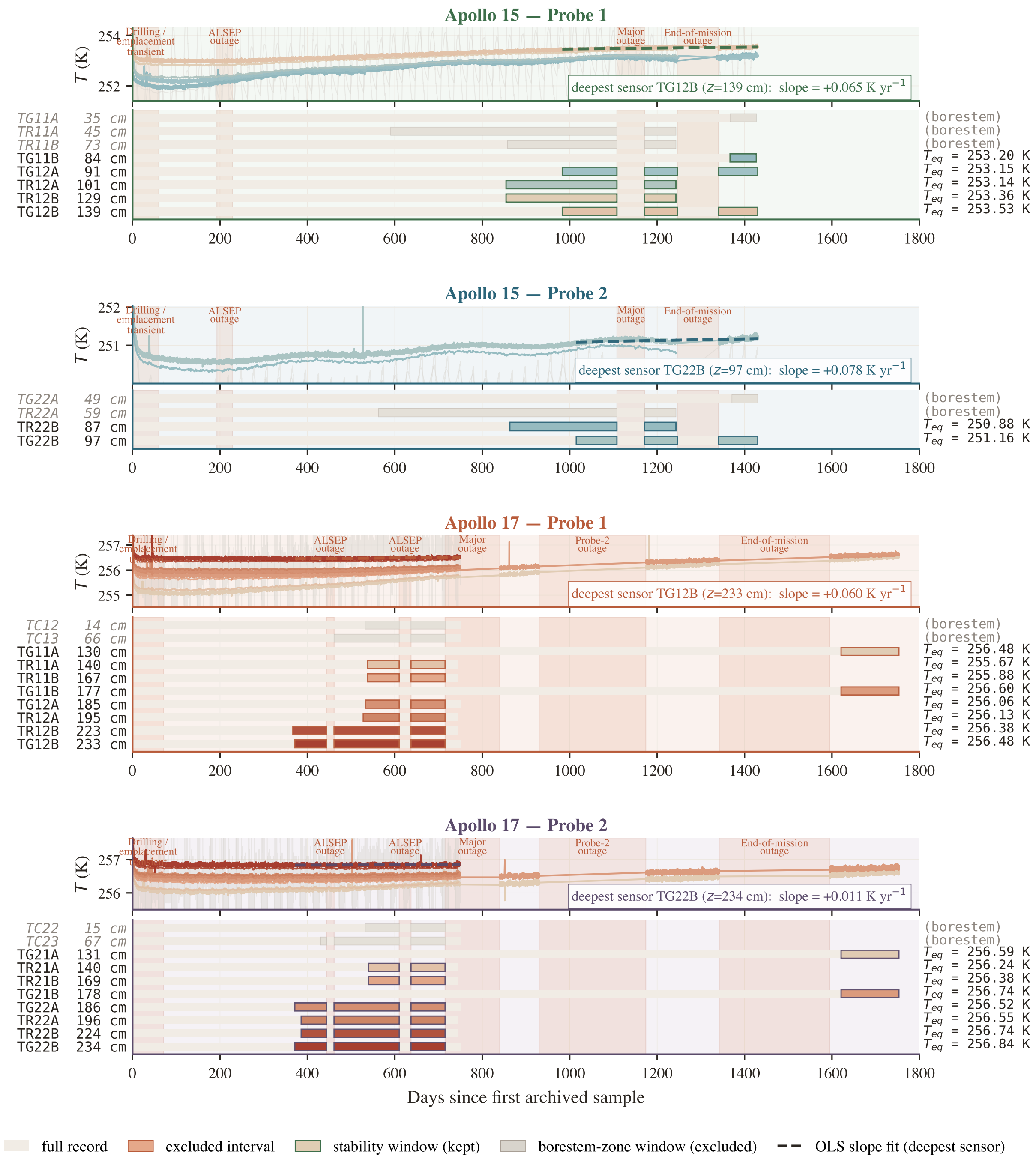

Figure 3. Per-probe stability-window timeline at Apollo 15 and 17 (4 probes). Each probe panel shows, above, the raw HFE traces with coral bands marking documented data-quality events and the selected stability windows outlined; and, below, a depth-coded per-sensor Gantt strip broken at outages. Selection criteria and the role of the window-mean $T_{\rm eq}$ are described in §2.

2.2 Surface solar geometry and incident insolation

The solver is forced with an idealized synodic insolation cycle: the daytime solar zenith angle at each site is approximated by $\cos\theta_\odot(t) = \cos\phi\,\cos(2\pi t/P)$, with ϕ the site latitude and $P = 29.531$ d the mean synodic period. Because the retrieval compares stability-window mean temperatures (§2) rather than time-resolved phases, ephemeris-level timing corrections do not enter the fit. The neglected solar declination ($\pm1.54°$) and orbital eccentricity ($\pm3.3\%$ peak-to-peak in instantaneous flux, $< 0.1\%$ in the cycle mean) perturb the mean insolation at or below the percent level, comparable to and subsumed by the Bond-albedo sensitivity of Table 4. The Moon has no atmosphere, so the solar irradiance incident on the local horizontal surface is

$$F_\odot(t) = \max(0,\ S_0 \cos\theta_\odot(t)), \tag{1}$$

with $S_0 = 1361\,\mathrm{W/m^2}$ (Kopp and Lean, 2011).

2.3 1-D solver and Hayne et al. (2017) thermophysics

The depth coordinate z is defined positive downward, with the regolith surface at $z = 0$ and the lower boundary of the modeled column at $z = z_{\max} = 5$ m. Conservation of energy in one dimension gives the heat equation

$$\rho(z)\, c_p(T)\, \frac{\partial T}{\partial t} = \frac{\partial}{\partial z}\Big[\, K(T,z)\, \frac{\partial T}{\partial z}\Big]. \tag{2}$$

At the upper boundary the skin temperature T_s satisfies radiative balance between absorbed insolation, emitted thermal radiation, and the conductive flux into the column,

$$(1-A)\, F_\odot(t) \;-\; \varepsilon\sigma T_s^4 \;-\; \big(-K\,\partial_z T\big)_{z=0} \;=\; 0, \tag{3}$$

with $\sigma = 5.6704\times10^{-8}$ W m^{-2} K^{-4} the Stefan–Boltzmann constant; equation (3) is solved iteratively for T_s at each time step. At the lower boundary a constant geothermal flux is imposed,

$$K\,\partial_z T\big|_{z_{\max}} = Q_b, \tag{4}$$

with z positive downward: the upward geothermal flux maintains a positive (warming-with-depth) temperature gradient, matching the solver's basal ghost condition. The PDE is discretized on a geometric grid ($\Delta z_0 = 2$ mm, growth ratio 1.08) and integrated with a Crank–Nicolson scheme at $\Delta t = 1800$ s on a lunation-commensurate time grid whose final (cycle-closing) step is marched explicitly, with material properties re-evaluated at the time-step midpoint by two corrector sweeps (the implicit-midpoint prescription; both details are required for the periodic state to converge at second order in Δt). The periodic steady state is computed by a flux-anchored outer iteration rather than a fixed-length spin-up: short multi-lunation integrations converge the diurnal skin, after which the sub-skin mean profile is reconstructed from the steady-state mean-flux closure $d\langle T\rangle/dz = [Q_b - u_{\mathrm{rect}}(z)]/K(\langle T\rangle, z)$, where the rectified (eddy-correlation) flux $u_{\mathrm{rect}} = \langle K\,\partial_z T\rangle - K(\langle T\rangle)\,\partial_z\langle T\rangle$ is measured from the resolved cycle; the two steps are alternated until the anchor temperature at $z = 0.55$ m is stable to 5×10^{-3} K. The skin integration is confined to a shallow sub-column ($z \le 0.70$ m) with Q_b imposed at its base, while the deep column is supplied entirely by the closure reconstruction and never time-stepped; this decouples the spin-up from the $\sim10^3$-lunation relaxation of the full 5-m domain, so a modest inner length is free of the initial condition. Convergence is judged on the retrieved K_d^* itself: it is stable to $< 5\times10^{-3}$ mW m^{-1} K^{-1} once $n_{\mathrm{inner}} \gtrsim 64$ at Apollo 17 and $\gtrsim 96$ at the lower-conductivity, slower-relaxing Apollo 15 site, which sets the production $n_{\mathrm{inner}} = 96$; the truncated and full-column constructions agree on K_d^* to 2×10^{-3} mW m^{-1} K^{-1}, confirming the truncation is unbiased. Initial temperature guesses of 240 and 260 K yield mean profiles agreeing to ≤ 0.03 K everywhere in the column, and grid-and-timestep refinement, anchor-depth variation over 0.45–0.70 m, and the K_d-grid spacing bound the residual solver systematic on K_d^* at $\lesssim 0.04$ mW m^{-1} K^{-1} — well below the bootstrap uncertainty (§3.2). Temporal convergence is certified in the reported quantity itself: halving Δt from 1800 to 900 to 450 s moves K_d^* by ≤ 0.018 (A17) and ≤ 0.001 (A15) mW m^{-1} K^{-1}, and the Richardson extrapolation of an intentionally first-order variant of the scheme converges to the same K_d^*(A17) to four significant digits — two independent discretizations agreeing on the continuum limit. The solver is further verified against an independent line-for-line C++ implementation, with which it agrees to 10^{-12} K over multi-lunation integrations; all certification artifacts are archived with the pipeline (§5). The Hayne et al. (2017) thermophysical inputs are adopted as given in Appendix A of the original paper:

$$K(T,z) = \Big\{K_s + (K_d - K_s)\big[1 - e^{-z/H}\big]\Big\} \cdot \Big[1 + \chi\big(T/T_{\mathrm{ref}}\big)^3\Big], \tag{5}$$

$$\rho(z) = \rho_d - (\rho_d - \rho_s)\, e^{-z/H}, \tag{6}$$

$$c_p(T) = c_0 + c_1 T + c_2 T^2 + c_3 T^3 + c_4 T^4. \tag{7}$$

Throughout, K_d denotes the deep asymptote of the *contact* component $K_c(z)$ of equation (5); the total conductivity $K(T, z)$ carries in addition the radiative factor $1 + \chi(T/T_{\rm ref})^3$, which at the retained-sensor temperatures (≈252–257 K) multiplies the contact term by ≈2.0. At the retained K_d^* the mean *effective* conductivity at the sensors is 9.3 (A15) and 14.6 (A17) mW m^{-1} K^{-1}; literature comparisons must respect this distinction (§4.3). Numerical values and sources for every coefficient, together with the site-specific measured quantities (latitude, albedo, emissivity, basal heat flux), are listed in Table 1. The basal heat flux reanalyses of Saito et al. (2007); Nagihara et al. (2018) are folded into the sensitivity sweep of §3.2.

Table 1. Fixed inputs to the 1-D heat solver. Three blocks: (A) the Hayne et al. (2017) conductivity-form coefficients (eq. 5), with K_d the single parameter retrieved in §2.5; (B) the four published coefficients of the Martínez and Siegler (2021) $K(T, \rho)$ comparison model (eq. 8, forward only in the headline table; see also §4.3); (C) site-specific measured quantities shared by both conductivity models. The shared $\rho(z)$ uses the Hayne-form profile (eq. 6); the specific heat is the Hemingway et al. (1973) polynomial $c_p(T) = c_0 + c_1T + c_2T^2 + c_3T^3 + c_4T^4$ with $(c_0, \ldots, c_4) = (-3.61,\ 2.74,\ 2.36\times10^{-3},\ -1.23\times10^{-5},\ 8.91\times10^{-9})$.

Parameter	Symbol	Value	Unit	Source
(A) Hayne et al. (2017) smooth-exponential conductivity, all values from Hayne et al. (2017) App. A unless noted				
Surface conductivity	K_s	7.4×10^{-4}	W m^{-1} K^{-1}	
Deep conductivity (global)	K_d (publ.)	3.4×10^{-3}	W m^{-1} K^{-1}	
Deep conductivity (per-site)	K_d^*	§3.2	W m^{-1} K^{-1}	**this work**
$K(z)$ transition scale	H	0.06	m	
Radiative coefficient	χ	2.7	—	
Radiative normalization	$T_{\rm ref}$	350	K	
(B) Martínez & Siegler (2021) $K(T, \rho)$ (forward only)*, coefficients from Martínez and Siegler (2021)*				
Contact, ρ-slope	A_1	5.082×10^{-6}	W m^{-1} K^{-1} (kg m^{-3})$^{-1}$	
Contact, intercept	A_2	-5.10×10^{-3}	W m^{-1} K^{-1}	
Radiative, ρ-slope	B_1	2.002×10^{-13}	W m^{-1} K^{-4} (kg m^{-3})$^{-1}$	
Radiative, intercept	B_2	-1.953×10^{-10}	W m^{-1} K^{-4}	
Shared by both conductivity models				
Surface bulk density	ρ_s	1100	kg m^{-3}	Hayne et al. (2017)
Deep bulk density	ρ_d	1800	kg m^{-3}	Hayne et al. (2017)
Specific heat	$c_p(T)$	polyn. (caption)	J kg^{-1} K^{-1}	Hemingway et al. (1973)
(C) Site-specific measured inputs (A15 / A17)				
Latitude	ϕ	26.1° / 20.2° N	—	ALSEP gazetteer
Bond albedo	A	0.131 / 0.137	—	Vasavada et al. (2012)
Broadband emissivity	ε	0.95	—	Bandfield et al. (2015)
Basal heat flux	Q_b	21 / 16	mW m^{-2}	Langseth et al. (1976); Nagihara et al. (2018)
Solar constant	S_0	1361	W m^{-2}	Kopp and Lean (2011)

As an independent comparison we also evaluate the Martínez and Siegler (2021) regolith conductivity model. Where Hayne et al. (2017) prescribes a depth-dependent envelope through $K(T, z) = [K_s + (K_d - K_s)(1 - e^{-z/H})](1 + \chi(T/T_{\rm ref})^3)$, Martínez and Siegler (2021) replaces that envelope with an explicitly density-dependent two-mechanism decomposition,

$$K(T, \rho) = \underbrace{(A_1\rho + A_2)\, k_{\rm am}(T)}_{\text{contact (phonon)}} + \underbrace{(B_1\rho + B_2)\, T^3}_{\text{radiative}}. \tag{8}$$

The first term is the contact (phonon) conductivity transmitted through the network of grain-to-grain contacts; it carries the temperature dependence of an amorphous solid through the low-temperature polynomial $k_{\rm am}(T)$ of Martínez and Siegler (2021) and scales linearly with the local bulk density, reflecting the increased number of grain-to-grain contacts per unit volume at higher packing. The second term is the radiative contribution from infrared photon exchange across pore voids; it scales as T^3 (the derivative of the Stefan–Boltzmann emission with respect to temperature, characteristic of photon-mediated transport) and is density-modulated with the opposite physical content, since denser packing reduces the void fraction available for radiative exchange. The four coefficients (A_1, A_2, B_1, B_2) are calibrated globally against Diviner brightness temperatures and are listed in Table 1. The density profile $\rho(z)$ used in this comparison is the same Hayne-form exponential used in the Hayne column, so the two models differ in their K formulation alone.

The Martínez and Siegler (2021) formulation has no single K_d analogous to the parameter retrieved under the Hayne form: both terms in equation (8) vary continuously with depth through $\rho(z)$ and with temperature through $T(z)$. The published model contains no free parameter, so its evaluation at each site yields a single forward prediction (Table 2). To diagnose, in the discussion (§4.3), whether the published Martínez and Siegler (2021) form could fit both sites if it were granted one per-site lever analogous to the Hayne K_d, we additionally perform an alternative retrieval in which the asymptote of the Hayne-form density profile entering equation (8) is rescaled by a single per-site scalar α,

$$\rho_d^{\text{site}} \;=\; \alpha \cdot 1800 \text{ kg m}^{-3}, \tag{9}$$

with $\rho_s = 1100$ kg m^{-3}, the e-folding depth H, and the published coefficients (A_1, A_2, B_1, B_2) and $k_{\text{am}}(T)$ held fixed; $\alpha = 1$ recovers the published forward prediction. The scalar α is our diagnostic construction (the Martínez and Siegler (2021) model itself is not refit, only re-evaluated at a per-site density), and its physical interpretation is the per-site asymptotic bulk density that the published $K(T, \rho)$ would require to reproduce the in-situ data. The α retrieval is not part of the headline K_d result; it is interpreted in §4.3.

2.4 Borestem diagnostic and the meter-scale-sensor cut

The meter-scale ($z > 80$ cm) sensor set used in the K_d retrieval is defined by a physics-based amplitude diagnostic, not by the residuals it would produce. The amplitude of the diurnal wave serves as a frequency-domain check independent of the lunation-mean comparison. The per-sensor diurnal amplitude is estimated as $\sqrt{2}\,\sigma_{\Delta t}$, where $\sigma_{\Delta t}$ is the within-stability-window standard deviation; the proxy equals the half-range exactly for a sinusoidal signal, is diurnal-dominated for shallow sensors, and saturates at the digitization noise floor below the skin depth — precisely the contrast the diagnostic requires (Fig. 4). Two features are diagnostic. Above $z = 80$ cm, observed amplitudes exceed the modeled regolith-attenuation curve by an order of magnitude or more, the signature of heat conducted along the fiberglass borestem from the lunar surface, which thermally short-circuits the upper ~ 80 cm of regolith (Nagihara et al., 2018). Below 80 cm, both published $K(z)$ forms predict diurnal amplitudes $\lesssim 10^{-7}$ K ($\sim$18 e-foldings of the diurnal skin depth $\delta \sim 3$–5 cm), two to three orders of magnitude below the 0.05–0.2 K digitization noise floor of the restored record. Meter-scale-sensor amplitudes therefore do not constrain the thermal diffusivity $\kappa = K/(\rho c_p)$ at the in-situ scale; the Hayne and Martínez and Siegler (2021) attenuation curves plotted in Figure 4 agree to within $\sim$10% in skin depth and overlap visually below the noise floor, both indicating "deep = unresolved". The shallow-sensor amplitude excess, evaluated against a prediction that depends only on $K(T, z)$, $\rho(z)$, and $c_p(T)$, defines the meter-scale ($z > 80$ cm) validation set used in the retrieval.

2.5 Per-site K_d retrieval (Hayne shape held fixed)

K_d is retrieved separately at each site by minimizing the meter-scale-sensor RMSE while holding the remainder of the Hayne functional form (eqs. 5–7) fixed. $(K_s, H, \chi, T_{\text{ref}})$ and the density and c_p functions are held at their published values, and K_d is swept across site-specific grids: 29 points spanning 1.0–15.0×10^{-3} W/(m K) at A15 and 32 points spanning 3.0–25.0×10^{-3} W/(m K) at A17, sized so that both parabolic minima and the bootstrap upper percentiles fall inside the swept range (Fig. 7). At each K_d a full Crank–Nicolson spin-up is run to convergence, and the meter-scale-sensor RMSE is evaluated as

$$\text{RMSE}(K_d) = \Big\langle \big[T_{\text{model}}(z_i; K_d) - T_{\text{eq}}^{\text{HFE}}(z_i)\big]^2 \Big\rangle^{1/2}_{z_i \geq 80\,\text{cm}}. \tag{10}$$

The point estimate K_d^* is the parabolic minimum through the three grid values bracketing the RMSE minimum. Its uncertainty is quantified by a non-parametric bootstrap with $N_{\text{boot}} = 1500$ resamples of the meter-scale-sensor set drawn with replacement, with the ± 2.5 cm sensor-placement

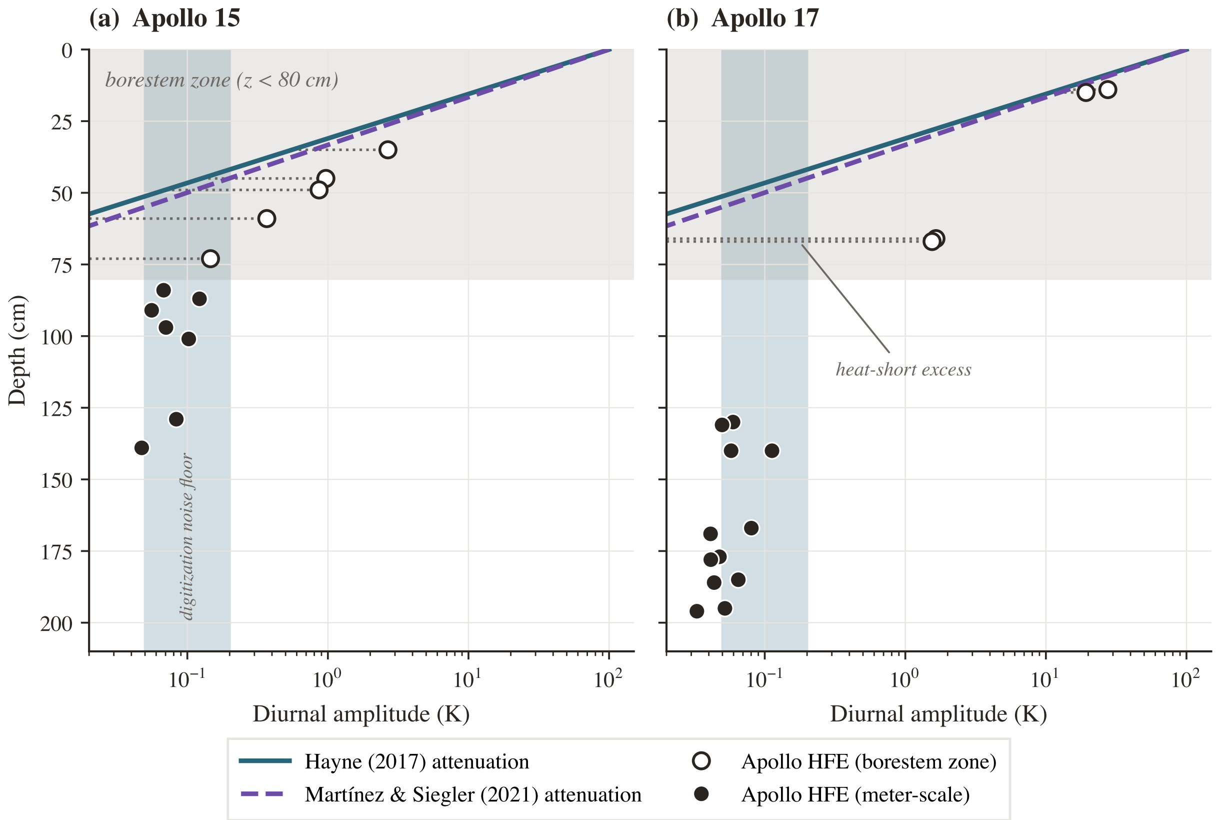

Figure 4. Diurnal amplitude ($\sqrt{2}\times$ the within-stability-window standard deviation) versus sensor depth at both Apollo sites. Open circles: borestem-zone sensors ($z < 80$ cm); filled circles: meter-scale sensors; dotted leaders mark each borestem sensor's heat-short excess above the modeled attenuation. Solid teal: Hayne et al. (2017) attenuation; dashed violet: Martínez and Siegler (2021) — visually indistinguishable on the log axis ($T = 250$ K, published density 1800 kg m^{-3}). The order-of-magnitude shallow-sensor excess is the borestem heat-short signature.

uncertainty (Nagihara et al., 2018) propagated as a per-resample Gaussian depth jitter, following standard small-sample regression practice (Press et al., 2007). The bootstrap distribution defines 95% confidence intervals on K_d^* and on the inter-site contrast ΔK_d^*, and provides the bootstrap tail proportion $P_{\mathrm{boot}}(\Delta K_d^* \leq 0)$, the fraction of fitted (non-null-centered) resample draws with a non-positive contrast, reported in place of a conventional null-centered p-value (Fig. 7). Percentile-method intervals are reported throughout; bias-corrected accelerated (BCa) intervals (Efron and Tibshirani, 1993) agree with the percentile endpoints to within 1.3 mW/(m K) at both sites and on the contrast, and do not alter any qualitative conclusion — a difference comparable to the CI half-widths themselves ($\sim$1.4 at A15, $\sim$1.0 at A17), so the percentile intervals should be read as approximate at these sample sizes ($N = 7$ and 16). The retrieval has one free parameter.

Models with different free-parameter counts on the same meter-scale-sensor set are compared using the small-sample-corrected Akaike information criterion (Burnham and Anderson, 2002), $\mathrm{AICc} = N \ln(\mathrm{RSS}/N) + 2k + 2k(k{+}1)/(N{-}k{-}1)$, where RSS is the meter-scale-sensor residual sum of squares and k counts the fitted parameters plus one for the residual-variance estimate ($k = 1$ for fixed-K_d rows, $k = 2$ for each retrieval). Differences ΔAICc are reported relative to the best model at each site; following Burnham and Anderson (2002), $\Delta\mathrm{AICc} \lesssim 2$ indicates statistically indistinguishable models and $\Delta\mathrm{AICc} \gtrsim 10$ a decisive preference. A reduced χ^2 is not reported: the within-stability-window scatter ($\sigma \sim 0.03$–0.05 K) characterizes short-term thermometer precision rather than the model-adequacy error budget, yields $\chi^2_\nu \gg 1$ for every model, and is therefore uninformative for model selection in this context.

3 Results

3.1 Annual-mean subsurface profile and the global-model gap

Both published global models reproduce the lunation-mean borehole profiles to within ∼1 K, but systematic per-site residuals remain. Figure 5 compares the two modeled time-averaged $\langle T(z)\rangle$ profiles, Hayne et al. (2017) at the global $K_d = 3.4\,\mathrm{mW/(m\,K)}$ and Martínez and Siegler (2021) $K(T,\rho)$ evaluated forward, against the stability-window means. The published Hayne value runs warm relative to the HFE record by a mean of +0.62 K at A15 and +0.54 K at A17, with RMSE 1.09 and 0.89 K — the residual structure, not the mean offset, carries the per-site information; the Martínez and Siegler (2021) prediction is comparable (RMSE 1.05 and 0.63 K) (Table 2). The shaded band ($z < 80\,\mathrm{cm}$) marks the borestem-contaminated zone, excluded *a priori* from the quantitative comparison (§2.4). The residual structure that remains — in particular the under-predicted temperature gradient between the 80-cm and deepest sensors at A17 — motivates the per-site K_d retrieval of §3.2.

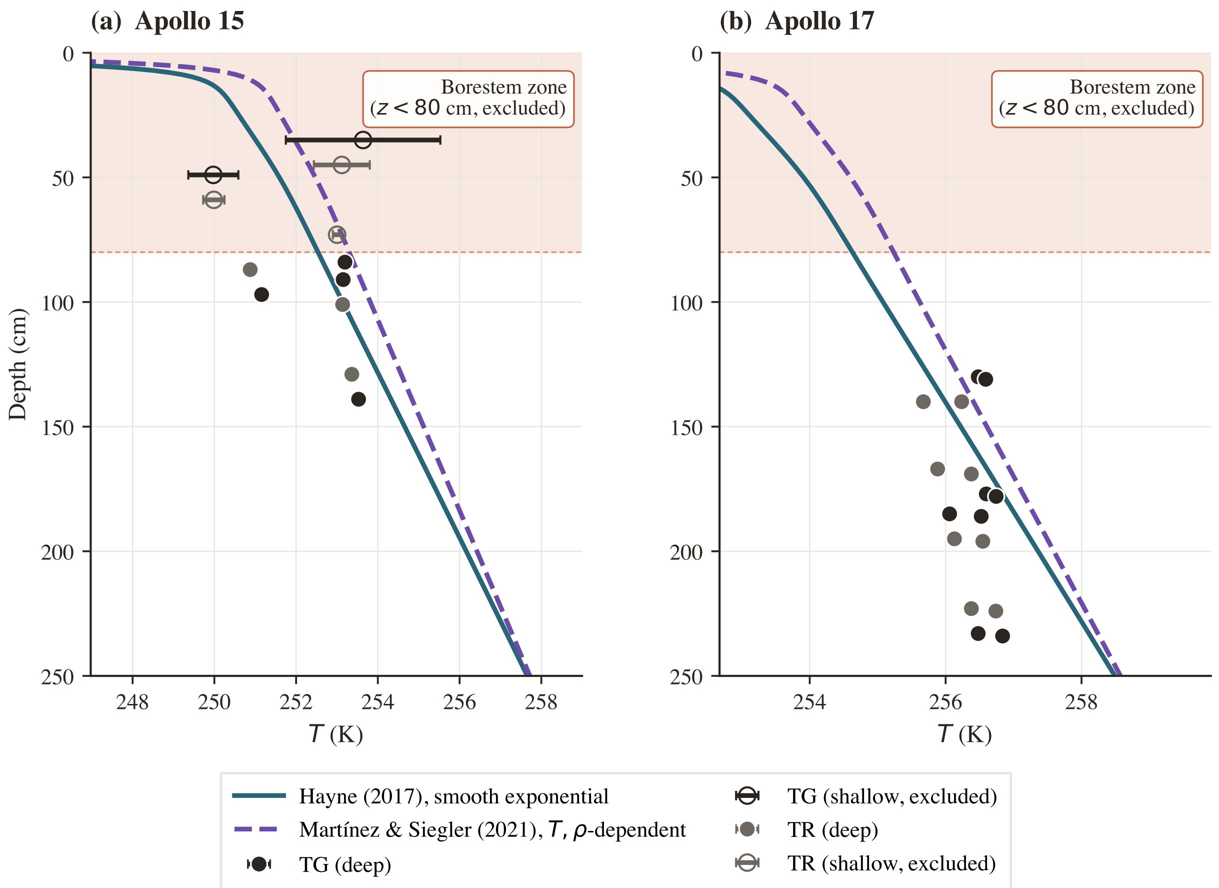

Figure 5. Annual-mean subsurface temperature $\langle T(z)\rangle$ for the two published models at nominal (un-retrieved) parameters — Hayne et al. (2017) at the global $K_d = 3.4\,\mathrm{mW/(m\,K)}$ (solid) and Martínez and Siegler (2021) $K(T,\rho)$ forward (dashed) — against the Apollo HFE stability-window means (charcoal = TG, gray = TR sensors; bars = within-window standard deviation). Shaded band: the borestem zone ($z < 80\,\mathrm{cm}$), excluded from the RMSE.

3.2 Per-site K_d retrieval

The meter-scale-sensor RMSE exhibits a well-resolved minimum at each site:

$$K^*_{d,\mathrm{A15}} = 4.60^{+0.86}_{-0.21}\,[4.18,\ 6.96]\,\mathrm{mW/(m\,K)}, \qquad K^*_{d,\mathrm{A17}} = 7.08^{+0.53}_{-0.52}\,[6.16,\ 8.07]\,\mathrm{mW/(m\,K)},$$

with asymmetric 1σ bootstrap uncertainties and 95% confidence intervals in brackets (Fig. 7). The meter-scale-sensor RMSE decreases from 1.09 K and 0.89 K under the published global K_d to 1.00

Table 2. Headline meter-scale-sensor RMSE table. Meter-scale-sensor (z = 80–234 cm) validation statistics for the three forward configurations confronted with the same Apollo HFE record: the Hayne et al. (2017) form at its published global K_d, the Hayne et al. (2017) form with K_d refit per site (the retrieval of §3.2), and the Martínez and Siegler (2021) $K(T, \rho)$ model evaluated forward at each site. K_d in mW m^{-1} K^{-1} (95% bootstrap CI in brackets for the Hayne site-fit row). Bias = mean(model − obs). A diagnostic per-site retrieval under the Martínez and Siegler (2021) form is presented separately in §4.3.

Site	Model	Fitted knob	N	RMSE (K)	Bias (K)
Apollo 15	Hayne (2017), global K_d	—	7	**1.09**	+0.62
Apollo 15	Hayne, K_d^* fit	$K_d^* = 4.60\,[4.18, 6.96]$	7	**1.00**	+0.44
Apollo 15	Martínez and Siegler (2021) forward	—	7	**1.05**	+0.54
Apollo 17	Hayne (2017), global K_d	—	16	**0.89**	+0.54
Apollo 17	Hayne, K_d^* fit	$K_d^* = 7.08\,[6.16, 8.07]$	16	**0.40**	+0.12
Apollo 17	Martínez and Siegler (2021) forward	—	16	**0.63**	+0.09

K and 0.40 K under the per-site retrieval ($N = 7$ at A15, $N = 16$ at A17; Table 2); at A17 the refit halves the RMSE and removes the gradient misfit (bias +0.12 K), while at A15 the improvement is modest and the global value is not excluded by the RMSE criterion alone — it is excluded by the bootstrap CI, whose lower bound (4.18) lies above 3.4. K_d is therefore the dominant identifiable lever between the Hayne form and the in-situ HFE record once the borestem zone is excluded, with the caveat that the per-site evidence is decisive at A17 (ΔAICc $\approx$ 23 against the global value) but not at A15 alone (ΔAICc $\approx$ 3 in favor of the global value once the parameter penalty is paid; the inter-site contrast, not the A15 refit, carries the statistical result.

The minimum is not an artifact of the sweep grid. Extending the sweep beyond the fitting envelope reveals that each RMSE curve rises monotonically on the high-K_d side (Fig. 6), confirming bracketed minima rather than descending shoulders. Nor is it an artifact of the thermal-state initialization: the equilibrium construction of §2.3 is certified independent of its initial guess to ≤ 0.03 K, and a grid-and-timestep convergence study (halving Δz_0 and Δt) bounds the solver contribution to K_d^* at $\lesssim 0.05\,\mathrm{mW/(m\,K)}$, an order of magnitude below the bootstrap uncertainty quoted above.

The Martínez and Siegler (2021) model, evaluated forward at each site with its published global coefficients, serves as the global-baseline comparison: meter-scale-sensor RMSE of 1.05 K at A15 and 0.63 K at A17, with mean biases of +0.54 K and +0.09 K (Table 2). It reproduces both sites to within ~1 K without any per-site freedom — comparable to the Hayne site-fit at A15, though still a factor of ~1.6 above it at A17. A diagnostic per-site retrieval under this form is presented in §4.3, where the implications for the $K(T, \rho)$ parameterization are discussed.

Sensitivity to the borestem exclusion depth. The retrieval is repeated with the borestem exclusion depth z_b swept over 60–90 cm. The retrieved values remain stable across all admissible cuts: $K^*_{d,\mathrm{A15}} = 4.6 - -4.9$ and $K^*_{d,\mathrm{A17}} = 7.2\,\mathrm{mW/(m\,K)}$ for z_b = 70, 80, and 90 cm, yielding a contrast of 2.6–$3.0\,\mathrm{mW/(m\,K)}$, unchanged within rounding from the nominal value. The 60 cm cut alone deviates ($K^*_{d,\mathrm{A17}} = 8.8$, contrast 3.9), because at this depth a sensor inside the borestem-affected zone re-enters the A17 set and inflates the retrieved gradient, precisely the contamination that the amplitude-based cut (§2.4) is designed to exclude. The retrieval is therefore insensitive to z_b across its physically defensible range, and the 60 cm outlier confirms rather than undermines the borestem diagnostic.

Two summary statistics must be distinguished. The K_d^* values quoted above are parabolic point estimates of the RMSE minimum; the bootstrap medians of the same quantities are 4.63 and $7.08\,\mathrm{mW/(m\,K)}$ (Table 4). The inter-site contrast is defined as the bootstrap median of the paired within-resample difference, $\Delta K_d^* = 2.31^{+0.68}_{-0.90}\,\mathrm{mW/(m\,K)}$ (1σ, asymmetric; 95% CI [−0.12, 3.56], $P_\mathrm{boot}(\Delta K_d^* \leq 0) \approx 0.031$; Fig. 7). The two sites share no sensors, so the contrast draws combine independent per-site resamples; an all-pairs convolution of the two bootstrap distributions reproduces the paired interval to rounding precision. The median-of-differences construction is also the reason 2.31 differs slightly from the difference of the point estimates ($7.08 - 4.60 = 2.48$). The A15 upper

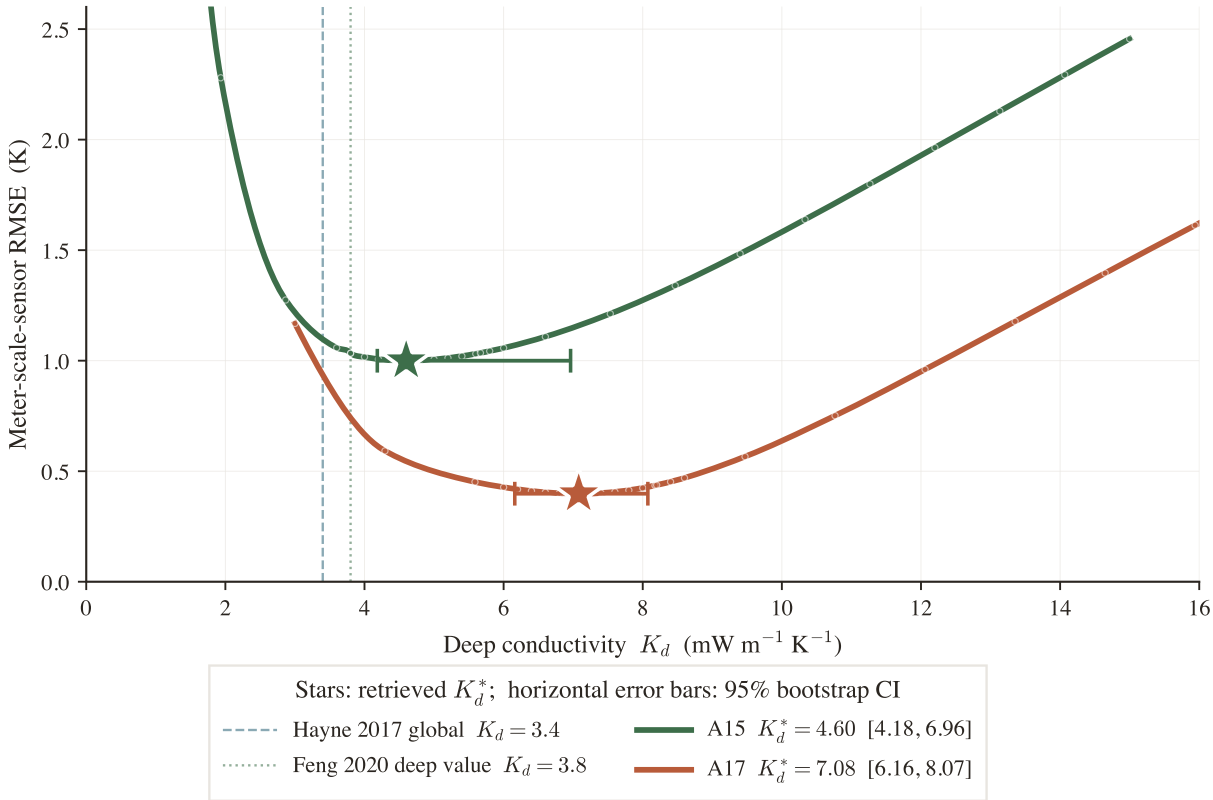

Figure 6. Per-site K_d retrieval under the Hayne et al. (2017) form (all other Hayne parameters at published values). Meter-scale-sensor RMSE(K_d) at each site, extended past the fitting grid so each curve rises again on the high-K_d side — the minima are genuine bracketed bowls. Stars: retrieved K_d^* (horizontal bars: 95% bootstrap CI); markers: forward-model evaluations. Dashed vertical: Hayne et al. (2017) global $K_d = 3.4$; dotted: Feng et al. (2020) deep value $K_d = 3.8$.

tail reflects a weakly constrained shoulder in RMSE(K_d) at A15 for $N = 7$ meter-scale sensors, even though the central parabolic minimum is firmly at 4.60. The reported 1σ intervals propagate sensor-set composition, the ± 2.5 cm depth-placement uncertainty (Nagihara et al., 2018), and the parabolic curve-fit error; component-by-component impacts of the remaining Hayne inputs are tabulated in Table 4. Both retrievals reject the global $K_d = 3.4\,\mathrm{mW/(m\,K)}$ of Hayne et al. (2017) at the 95% level (bootstrap CI lower bounds 4.18 and 6.16).

Applied at each site, the retrieved K_d^* values yield meter-scale-sensor profiles that track the HFE observations within the ~ 1 K spread of the data (Fig. 8). The Martínez and Siegler (2021) forward, plotted alongside, matches the Hayne site-fit RMSE at A15 to within ∼5% but remains a factor of ∼1.6 above it at A17.

A full per-component error budget for K_d^* is deferred to §4.2 (Table 4), where the basal-flux degeneracy that dominates the budget is discussed.

3.3 Consistency checks

Three diagnostics test the per-site K_d^* values. Their status as independent or internal is identified explicitly.

An independent Diviner surface-temperature closure test (Fig. 10, discussed in §4.3) shows that both conductivity models reproduce the canonical lunar diurnal shape, confirming that the diurnal forcing chain is correctly captured. Two further internal checks follow.

(i) TG vs. TR cross-prediction. Fitting K_d on the gradient-bridge sensors (TG) alone and predicting the ring-bridge (TR) sensors, and vice versa, yields held-out RMSEs within ∼0.11 K of the in-sample values at both sites. *(ii) Leave-one-deepest-out.* Withholding the deepest sensor ($z = 139$ cm at A15, $z = 234$ cm at A17) and refitting K_d on the remainder predicts the held-out temperature to within $|\Delta| = 0.37$ K and $|\Delta| = 0.25$ K respectively. Both internal checks confirm

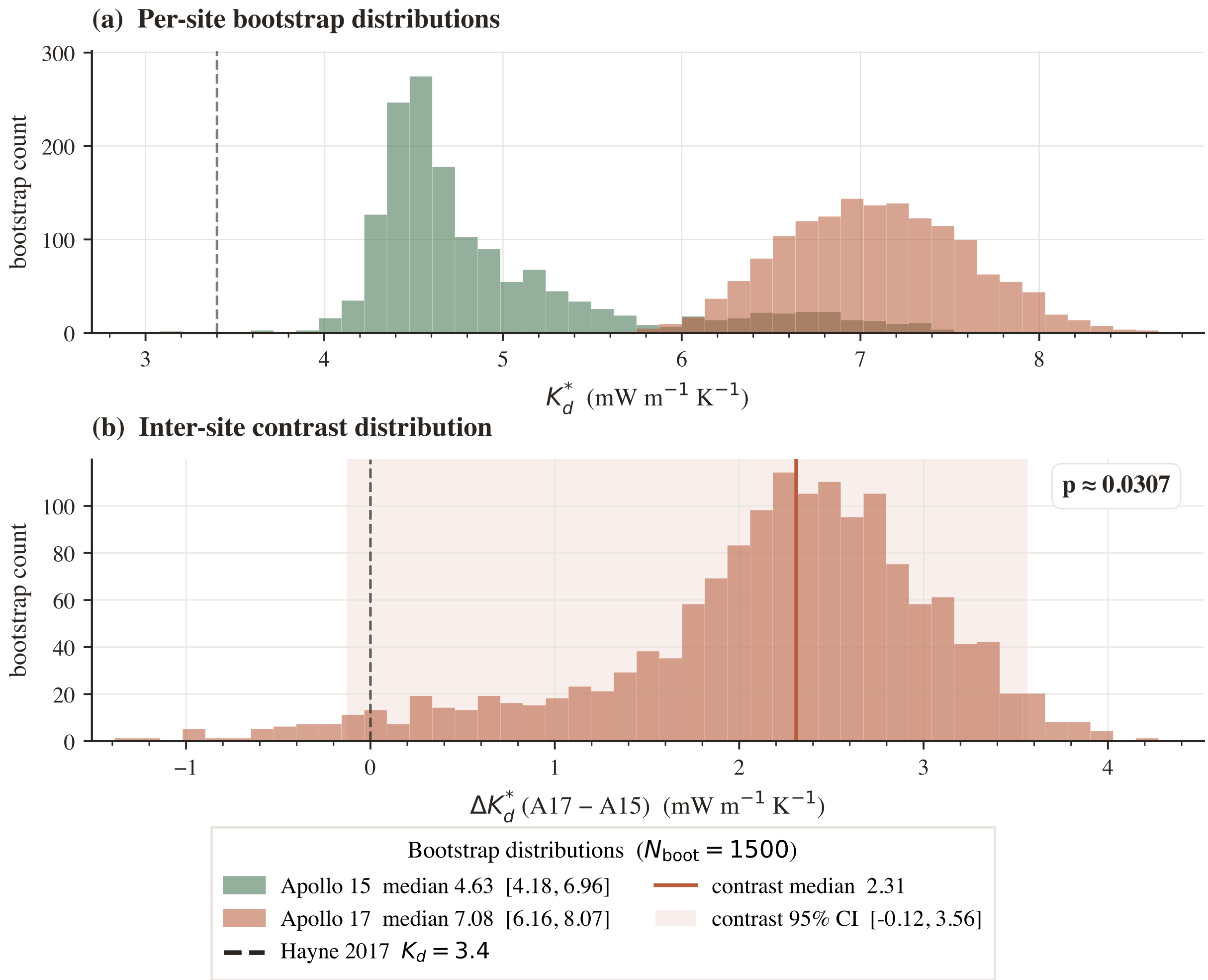

Figure 7. Bootstrap distributions ($N_{\rm boot} = 1500$, meter-scale-sensor resampling with ± 2.5 cm depth jitter). (a) Per-site K_d^* histograms (95% CIs in the legend); dashed vertical: the Hayne et al. (2017) global value. (b) Inter-site contrast $\Delta K_d^* = K_d^{\rm A17} - K_d^{\rm A15}$; 95% CI shaded, lower bound just below zero ($p \approx 0.031$).

that the retrieval is not driven by a single sensor or sensor type; neither constitutes an independent test of the inter-site contrast.

Bayesian cross-check. The non-parametric bootstrap is reported as the primary uncertainty estimate throughout because it makes no assumption about the basal heat flux and propagates only quantities measured at the boreholes (sensor-set composition and placement depth). As an independent cross-check on the *direction* of the contrast, not as a second estimate of its magnitude, we also ran a Bayesian inversion propagating the Saito et al. (2007); Nagihara et al. (2018) prior on Q_b (Fig. 9a). Its marginal MCMC medians are $K_{d,\rm A15} = 4.3$ and $K_{d,\rm A17} = 8.3$ mW m^{-1} K^{-1} (contrast median +3.9). The corresponding Q_b marginals pull below their priors at both sites — medians 17.8 (A15, prior 21 ± 3.2) and 13.9 mW m^{-2} (A17, prior 16 ± 2.4) — i.e., the HFE record itself mildly favors the lower, Saito et al. (2007)-direction fluxes, independently at the two boreholes; a firmer per-site Q_b remains the single most valuable external input this retrieval could receive. These differ from the bootstrap point estimates (4.60, 7.08; contrast +2.5) by construction, not by disagreement: the Q_b prior slides each marginal along the K_d/Q_b degeneracy ridge (§4.1), so the two methods are not independent measurements of the absolute K_d. The ordering not only survives the Q_b smearing — it is insensitive to it: on the directly solved RMSE(K_d, Q_b) surface (no proportional-degeneracy surrogate; §4.2), $P(K_d^{\rm A17} > K_d^{\rm A15})$ stays at 99.2–99.4% for Gaussian Q_b priors from $\pm 10\%$ to $\pm 40\%$ (archived scan: `results/qb_prior_width_scan.json`), and the contrast median grows with prior width (from +3.4 to +4.8 mW m^{-1} K^{-1}) because the two sites respond to a flux revision in opposite directions. The Bayesian and bootstrap views therefore bracket the honest statement: conditional on

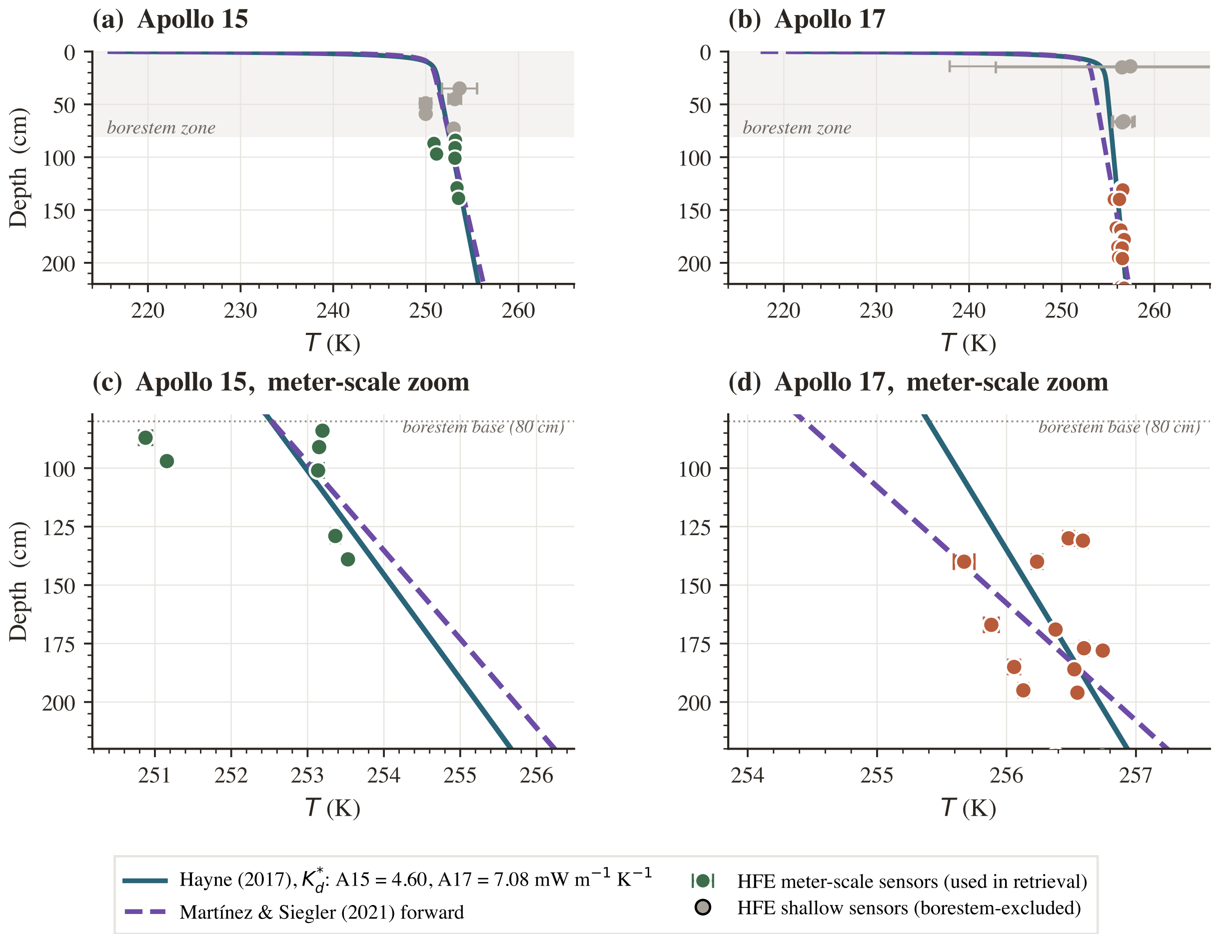

Figure 8. Annual-mean $T(z)$ at the two sites under the two models, against the Apollo HFE meter-scale sensors. Solid teal: Hayne et al. (2017) at the per-site retrieved K_d^*; dashed violet: Martínez and Siegler (2021) forward (no fitted knob). Top: full profile; bottom: meter-scale-sensor zoom. Colored markers: sensors used in retrieval; gray: borestem-affected shallow sensors (excluded).

the published basal fluxes the contrast magnitude is only marginally resolved ($P_{\rm boot}{\approx}0.03$, bootstrap 95% interval spanning zero), while once the basal-flux debate itself is propagated the *ordering* becomes decisive (99%, with the credible interval on the contrast excluding zero) — though its absolute magnitude then inherits the full flux uncertainty (§4.2). Both statements are conditional on the assumed per-sensor noise scale (0.5 K, chosen at the scale of the residual model–data RMSE of Table 2 rather than the 0.03–0.05 K within-window thermometer precision, which characterizes short-term stability only) that sets the likelihood sharpness.

Common-epoch sensitivity. The stability windows end at different calendar epochs while the record still warms: at A17 the retained sensors at 130–178 cm carry mid-1977 windows while every sensor below 180 cm carries a 1974 window (a 3.1-yr spread aligned with depth); at A15 the spread is 1.0 yr with trailing slopes $\leq$0.22 K yr^{-1}. Rebuilding every retained sensor's $T_{\rm eq}$ from one shared 1974 calendar window (every sensor has data there — no extrapolation) leaves A15 unchanged ($|\Delta K_d^*| \leq 0.01$) but moves A17 to $K_d^* = 6.16$ mW m^{-1} K^{-1} (-0.89; bootstrap 95% CI $[5.92, 6.81]$) — and improves the fit (RMSE 0.40 $\rightarrow$ 0.36 K), as expected if the mixed-epoch profile was internally inconsistent. A slope-translated construction agrees to 0.03 mW m^{-1} K^{-1}. The contrast under the common epoch is $+1.55$ mW m^{-1} K^{-1} (95% CI $[-0.11, 2.31]$, $P_{\rm boot}(\Delta K_d^*{\leq}0) = 0.04$): still positive, still rejecting the global value at both sites, but more marginal than at the certified windows. Epoch mixing is therefore the largest single methodological systematic at A17 and is carried as $\sigma_{\rm epoch}$ in Table 4 (archived: `results/common_epoch_sensitivity.json`).

The bootstrap interval on the contrast magnitude ($\Delta K_d^* = 2.3$, 95% CI $[-0.1, 3.6]$) is carried forward as the headline measurement.

4 Discussion

4.1 The two boreholes require different meter-scale temperature gradients

The HFE record does not admit a single meter-scale conductivity that fits both sites. Under the Hayne form at the published basal heat fluxes, the value that fits the A17 meter-scale-sensor residual (7.08) degrades A15 to a +0.70 K bias (RMSE 1.16 K), and the value that fits A15 (4.60) degrades the A17 RMSE from 0.40 to 0.54 K. The published global value of Hayne et al. (2017), applied unchanged, runs warm at both sites (+0.62 K at A15, +0.54 K at A17; Table 2) and, more importantly, under-predicts the A17 meter-scale temperature gradient. The per-site retrievals are $K_d^{\mathrm{A15}} = 4.60$ and $K_d^{\mathrm{A17}} = 7.08\,\mathrm{mW/(m\,K)}$ (95% bootstrap CIs [4.18, 6.96] and [6.16, 8.07]). The lower 95% bound on the contrast distribution (Fig. 7b) spans zero ($\Delta K_d^* = 2.31$, 95% CI $[-0.12, 3.56]$, $P_{\mathrm{boot}}(\Delta K_d^* \leq 0) \approx 0.031$): conditional on the published basal heat fluxes, the *requirement* of different meter-scale temperature gradients is established (decisively at A17, ΔAICc $\approx$ 23), while the *magnitude* of the conductivity contrast is favored by $\sim$97% of draws but not resolved at the 95% level. The conditionality matters — under the K_d/Q_b degeneracy of §4.2 the attribution to conductivity rather than basal flux still rests on the Q_b assumption.

The pooled three-model comparison (Table 3) has, at the certified numerics, no discriminating power at all: all three models — per-site K_d (M1), shared K_d with free per-site Q_b (M2), and the strict shared-K_d null (M3) — fall within ΔAICc $\leq$ 0.5 of one another, a statistical tie. This is expected rather than troubling: the pooled statistic dilutes the A17 signal with the less-informative A15 set ($N = 7$), whereas the per-site comparison is decisive at A17 (ΔAICc $\approx$ 23 for the site fit over the global value) and the bootstrap favors a positive contrast ($P_{\mathrm{boot}} \approx 0.03$; its 95% interval spans zero). The data therefore favor a per-site difference in meter-scale temperature gradient without excluding a uniform-conductivity alternative in which the two basal heat fluxes differ.

Table 3. Pooled meter-scale-sensor model comparison ($N = 23$: 7 at A15, 16 at A17). M1 retrieves a per-site K_d; M2 shares one K_d and frees the per-site Q_b within the published envelope; M3 fixes both. The three models are compared on a shared coarse (1 mW) K_d grid, so their tabulated K_d are rounded relative to the dense per-site headline retrieval (4.60/7.08); the pooled test probes model *structure*, not the vertex location. ΔAICc is relative to the best model. K_d in $\mathrm{mW\,m^{-1}\,K^{-1}}$, Q_b in $\mathrm{mW\,m^{-2}}$.

Model	K_d (A15 / A17)	Q_b (A15 / A17)	RMSE (K)	ΔAICc
M1 variable K_d, Q_b fixed	5.0 / 7.0	21 / 16 (fixed)	0.64	+0.1
M2 uniform K_d, Q_b free	7.0 (shared)	14 / 15 (fitted)	0.61	+0.5
M3 uniform K_d, Q_b fixed	6.0 (shared)	21 / 16 (fixed)	0.68	**0.0**

4.2 The retrieved absolute K_d values are degeneracy-limited

The absolute K_d^* values must be interpreted as conditional on the adopted basal heat flux, not as free-standing measurements. At leading order the meter-scale temperature gradient $|\partial_z T| = Q_b/K_d$ suggests the familiar proportional degeneracy $(K_d, Q_b) \to (\alpha K_d, \alpha Q_b)$; the retrieval, however, also matches the *absolute* sensor temperatures, whose offset through the low-conductivity skin column depends on Q_b/K non-proportionally — a direct test (rescaling Q_b and K_d together by 10/16 at A17 shifts the whole sensor band by a depth-uniform -1.3 K) shows the invariance fails, so the realized trade-off must be mapped by re-retrieval rather than assumed. The degeneracy is mapped quantitatively in Fig. 9a and is visible in the joint Bayesian posteriors. Adopting the canonical $Q_b^{\mathrm{A17}} = 16\,\mathrm{mW/m^2}$ (Langseth et al., 1976; Nagihara et al., 2018) yields $K_d = 7.08$; re-retrieving at the lower $\sim$10 mW m^{-2} value of Saito et al. (2007); Nagihara et al. (2018) moves the minimum up to $\sim$10.1 mW m^{-1} K^{-1} — and deepens it (RMSE 0.29 vs 0.40 K) — while the upper bound $\sim$18 mW m^{-2} lowers it to $\sim$6.8 mW m^{-1} K^{-1}. The A17 K_d^* therefore carries a $\sim^{+43}_{-4}\%$ envelope determined entirely by the basal-flux debate. A15 responds in the opposite sense — its steeper gradient dominates the fit, so K_d^* falls from 5.4 to 2.7 as Q_b^{A15} is revised from +25% to -25% of the published 21 mW m^{-2}. Because the two sites respond oppositely, the inter-site *contrast* is more robust to the flux debate than either absolute value: it remains positive across the entire debated envelope even with the two fluxes varied independently (worst pairing, both fluxes at their upper bounds: $6.8 - 5.3 = +1.5$ mW m^{-1} K^{-1}). What the boreholes cannot deliver is each site's *absolute* K_d — and fit quality alone cannot arbitrate the flux debate, since at A17 the Saito et al. (2007) flux actually fits the HFE record slightly better than the published value. This represents the intrinsic limit of what two boreholes can deliver.

Table 4. Per-component sensitivity budget for K_d^*. Entries are half-range sensitivity indices from scenario sweeps, not calibrated Gaussian standard deviations; the quadrature total is an independence-idealized summary (see text). Values in mW m^{-1} K^{-1}; entries below 0.05 are reported as <0.05. All components are re-derived with the certified equilibrium solver. The quadrature total covers the statistical and measured-input components; χ and H parameterize the Hayne et al. (2017) functional form that the retrieval holds fixed, and are reported separately as conditionality terms (see text) rather than summed — re-retrieving at the Vasavada et al. (2012) normalization $\chi = 1.48$ drives the RMSE minimum outside the physical sweep range at both sites.

Component	Envelope	Method	A15	A17
Statistical and solver				
σ_{stat}	± 2.5 cm depth jitter	bootstrap 16–84 % half-range	0.53	0.52
σ_{solver}	grid & Δt halved	convergence study, re-retrieve K_d^* (§2.3)	<0.05	<0.05
Measured inputs				
σ_{Q_b}	Q_b A15$\in[14, 25]$, A17$\in[10, 18]$	direct $K_d^*(Q_b)$ re-retrievals, half-range (§4.2; asymmetric: $^{+0.7}_{-2.5}$ / $^{+3.1}_{-0.3}$)	1.61	1.66
σ_A (albedo)	$A \pm 0.01$	re-retrieve K_d^* at perturbed A	0.72	2.96
σ_{K_s}	$K_s \pm 30\%$	re-retrieve at perturbed K_s	0.36	1.55
σ_ρ	$\rho_d \in [1700, 2000]$ kg m^{-3}	re-retrieve at perturbed ρ_d (ρ enters ρc_p only)	<0.05	0.33
Methodological choices (this work)				
σ_{thr}	0.04–0.16 K/yr (4× adopted)	threshold sweep, half-range of K_d^*	<0.05	<0.05
σ_{z_b}	$z_b \in \{70, 80, 90\}$ cm	cut sweep, half-range of K_d^*	0.12	<0.05
σ_{epoch}	common-1974 vs certified windows	re-fit at common-epoch T_{eq} (§3.3; one-sided: A17 -0.89)	<0.05	0.89
Total (quadrature)			**1.88**	**3.88**
Bootstrap median K_d^*			4.63	7.08
Conditionality of the held-fixed form (reported, not summed)				
χ	$\chi \in \{1.48, 2.7\}$	re-retrieve at the Vasavada et al. (2012) endpoint	minimum exits sweep (≤ 1.0 / ≥ 25)	
H	$H \in [3, 10]$ cm	per-H K_d^* half-range along the valley	0.51	3.47

Reading the per-component error budget. The derivation of each row in Table 4 falls into one of three categories: σ_{stat} is the 16–84 percentile half-width of the bootstrap distribution under the ± 2.5 cm astronaut-emplacement envelope reported by Nagihara et al. (2018); σ_{Q_b} is obtained in closed form through the K_d/Q_b degeneracy across the published reduction range (Langseth et al., 1976 canonical; Saito et al., 2007; Nagihara et al., 2018 reanalysis); every remaining row is computed by re-running the full retrieval at the perturbed input, with the envelopes taken from the published literature (σ_A: the Vasavada et al. (2012) lat-band albedo scatter of ± 0.01; σ_{K_s}: the Cremers and Birkebak (1971) laboratory scatter on Apollo 12 fines; σ_ρ: the Mitchell et al. (1973); Carrier et al. (1991) Apollo-core compaction range; σ_{solver}: the certified residual of the equilibrium construction, §2.3).

How the total is combined. The **Total (quadrature)** row is the independent-errors combination $\sigma_{\text{tot}} = \sqrt{\sum_i \sigma_i^2}$ over the statistical, solver, measured-input, and methodological rows. The worked arithmetic is

$$\sigma_{\text{tot}}^{\text{A15}} = \left(0.53^2 + 0.04^2 + 1.61^2 + 0.72^2 + 0.36^2 + 0.01^2 + 0.01^2 + 0.12^2\right)^{1/2} = 1.88,$$

$$\sigma_{\rm tot}^{\rm A17} = (0.52^2 + 0.04^2 + 1.66^2 + 2.96^2 + 1.55^2 + 0.33^2 + 0.03^2 + 0^2 + 0.89^2)^{1/2} \approx 3.88 \text{ mW m}^{-1}\,\text{K}^{-1}.$$

We flag one residual correlation: $\sigma_{\rm stat}$ and σ_{Q_b} both propagate along the K_d/Q_b degeneracy ridge of Fig. 9a, so the totals are independence-idealized rather than fully decorrelated standard errors; more generally, several rows are half-ranges of strongly asymmetric or nonlinear responses (the A17 albedo row especially), so the totals are sensitivity summaries, not calibrated 1σ uncertainties.

The conditionality terms. The χ and H rows are deliberately excluded from the quadrature: they are not measured inputs with Gaussian envelopes but coefficients of the Hayne et al. (2017) functional form that the retrieval holds fixed. Re-retrieving at the Vasavada et al. (2012) normalization $\chi = 1.48$ drives the RMSE minimum outside the physical sweep range at both sites — under the converged thermal physics, the rectified lunation-mean temperature at sensor depths depends strongly on the radiative coefficient, so the Hayne form at the Vasavada χ cannot reproduce the HFE record at any admissible K_d. K_d^* is therefore reported conditional on the published $(\chi, H) = (2.7, 6\,\text{cm})$, in precisely the same sense as it is conditional on the published Q_b; the inter-site *contrast* is far less exposed, since both sites share the same form.

Three conclusions follow. First, both budgets are now led by a single row: at A15 the direct-map σ_{Q_b} alone carries $\sim$73% of the variance (the flux debate, not the sensors, is the accuracy limit), whereas the A17 budget is dominated by σ_A and σ_{K_s}: at A17 the lunation-mean response to K_d is nearly flat over part of the sweep range, so even the ± 0.01 albedo envelope moves the minimum substantially — the statistical precision of the A17 retrieval (CI half-width $\sim$1.0) is real, but its accuracy is systematics-limited at the ± 3.8 level (the quadrature total less its statistical row). Second, σ_ρ is structurally suppressed under the Hayne $K(T, z)$ form (eq. 5), which contains no explicit ρ dependence: $\rho(z)$ enters only the volumetric heat capacity ρc_p and perturbs the rectified lunation-mean column at the $\lesssim$0.4 mW m^{-1} K^{-1} level. Third, the methodological-choice rows $\sigma_{\rm thr}$ and σ_{z_b} sit one to two orders of magnitude below the data-side terms, indicating robustness against both discretionary cuts; the exception is $\sigma_{\rm epoch}$ at A17, the epoch-mixing systematic quantified in §3.3, which rivals the statistical and K_s rows.

Behavior of the inter-site contrast under independent Q_b revision. How the contrast responds to a basal-flux revision cannot be read off a proportional-rescaling rule — the $(\alpha K_d, \alpha Q_b)$ invariance fails for this retrieval (§4.2) — so we map it explicitly by re-retrieving K_d^* at each site across the full adopted reanalysis envelope ($Q_{b,\rm A15} \in [14, 25]$, $Q_{b,\rm A17} \in [10, 18]$ mW m^{-2}; Fig. 9a, one direct re-retrieval per grid point). Because the two sites respond in opposite directions (A15's K_d^* rises with its flux, A17's falls), the corners of the envelope reorganize relative to the proportional picture. The favorable corner is both fluxes at their lower bounds ($Q_{b,\rm A15} = 14$, $Q_{b,\rm A17} = 10$ mW m^{-2}), where the contrast expands to +8.1 mW m^{-1} K^{-1} ($\sim 8.6\sigma$); the most adverse corner is both at their upper bounds (25, 18), where it compresses to +1.5 ($\sim 1.6\sigma$) — reduced, but not erased. A -33% revision of $Q_{b,\rm A17}$ alone raises the contrast to $\sim$+5.2. The conductivity contrast is therefore only marginally resolved at the published basal fluxes ($p \approx 0.03$, its 95% interval spanning zero), but its sign survives every admissible independent basal-flux assignment in the published envelope: A17 remains the more conductive site at every grid point of the direct map.

Contrast under the remaining leading systematics. The same direct-map logic extends to the two systematics that rival σ_{Q_b} in Table 4. The sites again respond in opposite directions: raising the Bond albedo by 0.01 lowers K_d^* at A15 (4.60→3.20) but raises it at A17 (7.08→11.81), so the contrast remains positive across the full ± 0.01 envelope even with the two albedos perturbed independently (most adverse pairing +1.3 mW m^{-1} K^{-1}); the ± 30% K_s envelope behaves identically (most adverse pairing +1.6). Like the basal-flux debate, these systematics limit each site's *absolute* K_d^* without threatening the sign of the contrast.

A joint (K_d, H) fit over the swept range H = 3–10 cm (Fig. 9b,c) confirms that H is effectively unconstrained: K_d and H trade off along a shallow valley: at A15 the interior minimum, $(4.3\,\text{mW m}^{-1}\,\text{K}^{-1}, 8\,\text{cm})$, improves on the 1-parameter retrieval at the published $H = 6$ cm by only 0.10 K of RMSE; at A17 the interior minimum, $(11.3\,\text{mW m}^{-1}\,\text{K}^{-1}, 8\,\text{cm})$, improves on the $H = 6$ slice by only 0.09 K. The reported K_d^* is therefore conditional on the published H, and the K_d–H

trade-off is propagated into the error budget as σ_H (Table 4). K_d, not H, is the identified parameter at fixed H; the pair is jointly identified only up to the valley direction.

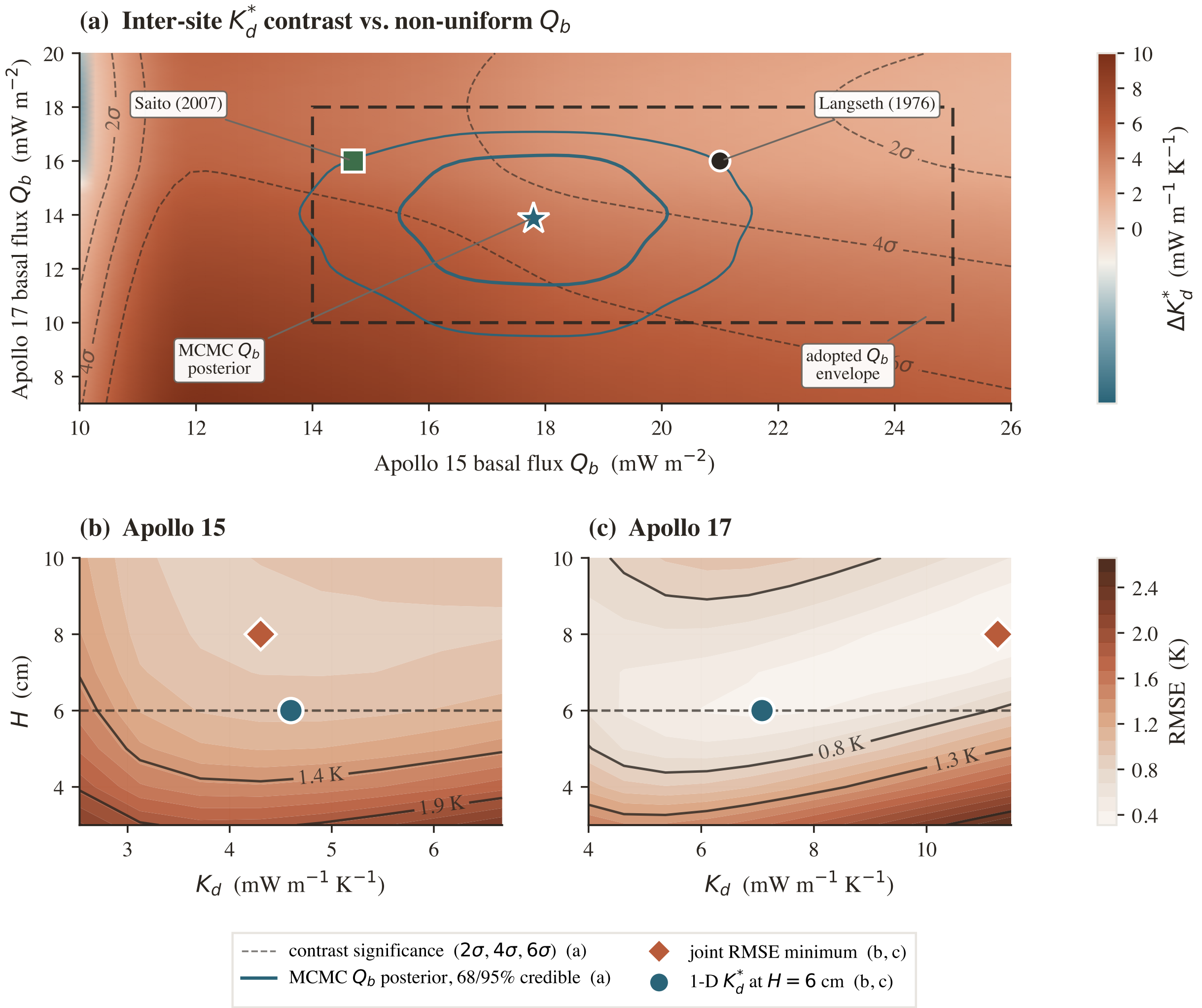

Figure 9. Robustness of the per-site K_d^* contrast. **(a)** Contrast ΔK_d^* across the two basal fluxes, one direct re-retrieval per grid point (no proportional-degeneracy assumption; §4.2); key features are labelled on the map and the dashed contours are bootstrap-σ significance. The MCMC posterior pulls both fluxes below their published values, yet the contrast stays positive throughout the adopted Q_b envelope (minimum $+1.5$ mW m^{-1} K^{-1}; it dips negative only in the far, implausible low-$Q_{b,\mathrm{A15}}$ corner), and the ordering $K_d^*(\mathrm{A17}) > K_d^*(\mathrm{A15})$ holds at ≈99.2%. **(b,c)** Joint (K_d, H) RMSE surface per site; the A17 sweep, now extended to 12 mW m^{-1} K^{-1}, reaches an interior valley bottom near 11.3 mW m^{-1} K^{-1} at $H \approx 8$ cm (coral diamond). Teal circle: 1-parameter retrieval at $H = 6$ cm.

4.3 Comparison with Martínez & Siegler (2021) and prior estimates

The second globally calibrated form fares notably better. The Martínez and Siegler (2021) $K(T, \rho)$ model, with its published coefficients and no per-site freedom, yields a meter-scale-sensor RMSE of 0.99 K at A15 and 0.64 K at A17 (Table 2, Fig. 8) — comparable to the Hayne site-fit at A15 and within a factor of ∼1.9 of it at A17. Its explicitly density- and temperature-dependent decomposition evidently transfers from the diurnal-skin calibration to the meter-scale regolith more faithfully than the Hayne smooth-exponential at its global K_d. The per-site Hayne retrieval nonetheless remains the sharper instrument at A17, where it halves the residual RMSE.

Diagnostic: Martinez per-site density retrieval. The fairness of the forward Martinez comparison can be tested by granting the published $K(T, \rho)$ one per-site lever and asking whether that lever can absorb the A17 residual within physical bounds. We note at the outset that Martínez and Siegler (2021) calibrated their four coefficients against surface brightness-temperature observations and did not validate the extrapolation to depths of order one meter; the test we run here is therefore a check of whether the published surface-calibrated form, when extended to the meter-scale regolith, can fit the in-situ HFE data with one per-site lever (the asymptotic bulk density entering the model). We do so under the construction of §2.5 (eq. 9): rescale the asymptote of the Hayne-form $\rho(z)$ entering equation (8) by a single per-site scalar α, leaving the four Martinez coefficients fixed, and sweep α against the same meter-scale-sensor RMSE. The retrieved values are $\alpha^*_{\mathrm{A15}} = 1.19$ (asymptotic bulk density $\rho^*_d \approx 2139$ kg m^{-3}, RMSE 0.91 K) and $\alpha^*_{\mathrm{A17}} = 1.04$ ($\rho^*_d \approx 1877$ kg m^{-3}, RMSE 0.60 K); at A15 the density lever slightly betters the Hayne site-fit RMSE (0.91 vs 1.00 K), while at A17 the bowl bottoms near the Hayne et al. (2017) density and the lever buys little ($0.63 \to 0.60$ K). The physical interpretation of α is the asymptotic bulk density that the published Martinez $K(T, \rho)$ would require to reproduce each site's in-situ profile.

The A15 retrieval sits ∼7% above the Apollo-core compaction ceiling of ∼2000 kg m^{-3} (Mitchell et al., 1973; Carrier et al., 1991) — outside the core envelope, though far below any solid-rock bound, and best read as marginal rather than comfortable. The A17 retrieval bottoms at $\rho^*_d \approx 1877$ kg m^{-3}, comfortably inside the Apollo-core envelope (1700–2000 kg m^{-3}): the published $K(T, \rho)$ is physically admissible at both boreholes. Figure S1 (Supporting Information) visualizes the result — both RMSE(α) bowls bottom inside or at the edge of the admissible band. The flat A17 bowl also exposes the limitation: because the bowl bottoms near the published density, no admissible density adjustment closes the remaining factor-of-∼1.5 RMSE gap to the per-site Hayne fit; sharpening the A17 gradient further requires freedom in the conductivity mechanism itself (the contact/radiative split), not in the density input. A full per-site recalibration of the Martínez and Siegler (2021) coefficients against the HFE record is the natural follow-on study.

The retrieved values reflect the steady-state model/data distinction. The solver represents the asymptotic regolith column that would exist under repeated solar forcing and the basal heat flux; the Apollo late tails are imperfect observations of that same asymptote because the probes were still recovering from emplacement. Residual drift can bias individual T_{eq} estimates by fractions of a kelvin, so it belongs in the systematic error budget, but it does not require an unphysical conductivity: the retrieved K_d values remain in the milliwatt-per-meter-per-kelvin range expected for porous lunar regolith.

Surface-temperature consistency check. The Diviner GCP is external to the retrieval (it enters neither the bootstrap nor the meter-scale-sensor RMSE), so it provides an independent check on the diurnal forcing chain (Fig. 10). Both parameterizations reproduce the canonical lunar diurnal shape, with full-cycle RMSE of 4.1 K at A15 and 6.3 K at A17 for the Hayne site fit, and 3.6 K at both sites for the Martinez forward. Both biases are dominated by the anisothermal-footprint and sub-pixel rock-population effects inherent to Diviner brightness temperatures (Bandfield et al., 2015; Hayne et al., 2017). Insolation, albedo, emissivity, and the upper-80 cm $K(T, z)$ are therefore correctly captured by both models, and the meter-scale-sensor discrepancy in Table 2 is not a surface-driven artifact.

At A17 the Hayne site fit runs surface-warmer than Diviner by +4.1 K while the Martinez forward sits at −0.8 K. The mechanism is direct: the high meter-scale $K^*_d = 7.08$ couples more heat upward through the diurnal skin layer than the globally calibrated $K(T, z)$ assumes, raising the dayside response. The per-site Hayne fit absorbs this penalty at the surface to fit the meter-scale temperature gradient; the surface-calibrated Martinez form remains essentially unbiased at the surface and pays a factor-of-∼1.6 penalty in the meter-scale-sensor RMSE relative to the Hayne site fit (Table 2). Within the Hayne $K(T, z)$ family, no single parameter choice simultaneously optimises the surface-brightness constraint of Williams et al. (2017) and the meter-scale-gradient constraint of Nagihara et al. (2018) at A17; the $K(T, \rho)$ family narrows but does not close that gap. The inter-site difference flagged by the meter-scale retrieval reappears in the surface-temperature data, in the form of which model fits which site at the surface and by how much.

4.4 Physical plausibility of the retrieved K^*_d

Apollo 15. The retrieval gives $K^{\mathrm{A15}}_d = 4.60\,\mathrm{mW/(m\,K)}$, above the global Hayne et al. (2017) value (3.4) and within the range expected for porous lunar regolith.

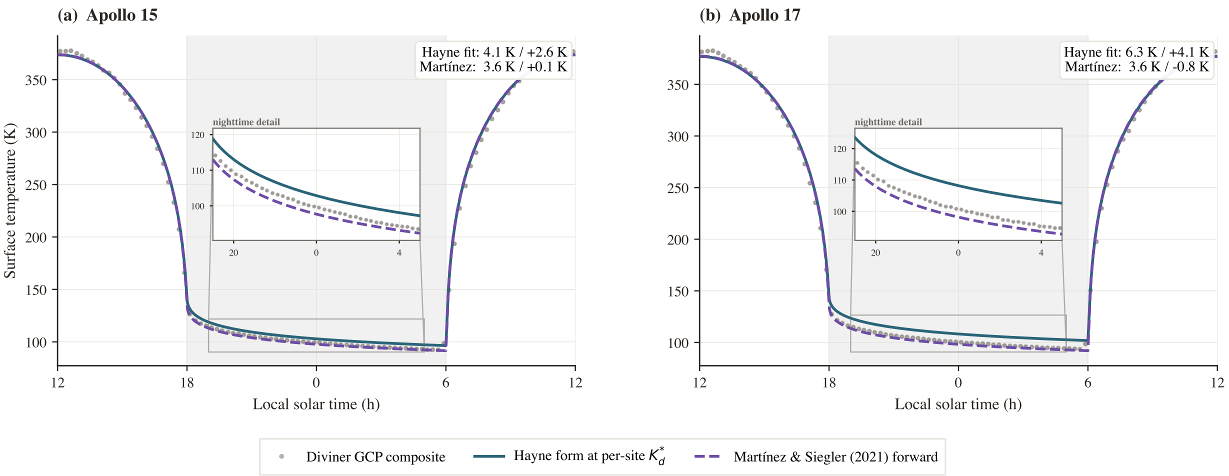

Figure 10. Surface-T closure against the Williams et al. (2017) Diviner GCP diurnal composite (gray markers). Teal solid: Hayne form at the per-site retrieved K_d^*; violet dashed: Martínez and Siegler (2021) at published coefficients. Local-solar-time axis centered on midnight (shaded band LST 18–06 = night); full-cycle RMSE / mean bias per panel in the inset.

Apollo 17. The retrieval gives $K_d^{\rm A17} = 7.08\,\mathrm{mW/(m\,K)}$, roughly twice the global Hayne value and 1.5 times the A15 retrieval, and likewise within the porous-regolith range.

Comparison with the in-situ reductions. Evaluated at the retrieved K_d^*, the model's effective conductivity at the retained sensors is 9.3 (A15) and 14.6 (A17) $\mathrm{mW\,m^{-1}\,K^{-1}}$. These are consistent with the revised annual-wave in-situ conductivities of Langseth et al. (1976) (their Fig. 6: ≈10 at A15 and ≈12 at A17 $\mathrm{mW\,m^{-1}\,K^{-1}}$, from the Table 1 diffusivities combined with the Hemingway et al. (1973) regolith heat capacity): the A15 value agrees to ∼7%, while A17 (14.6) sits at the upper edge of the in-situ range (≈11–14, its probe-to-probe and density scatter). The present work therefore re-parameterizes, rather than revises, the historical in-situ conductivity. Figure 11 places these values in their five-decade context, keyed by definition class.

A complementary test against the Martínez and Siegler (2021) $K(T, \rho)$ form (§4.3, Fig. S1, Supporting Information) is consistent: the density that form requires at A17, $\rho_d^* \approx 1877$ kg m^{-3}, lies inside the Apollo-core envelope, independently corroborating that the A17 meter-scale column is unremarkable in density and that the enhanced conductivity is carried by composition and contact structure rather than by anomalous compaction.

The A17 absolute value is conditional on the adopted basal heat flux. The Saito et al. (2007); Nagihara et al. (2018) reanalyses revise $Q_b^{\rm A17}$ downward by up to ∼30%; on the direct map (§4.2) a lower assumed flux moves the retrieved K_d^* up, to ∼10.1 mW m^{-1} K^{-1} (and fits the record slightly better), while the upper bound lowers it to ∼6.8. The flux debate therefore bounds K_d^*(A17) between ∼6.8 and ∼10.1 — all well above the global value. The factor-of-∼2 excess over the global Hayne value therefore cannot be read as a measured absolute conductivity independent of the Q_b assumption.

A site–depth ambiguity also deserves explicit mention: the two sensor sets barely overlap in depth (80–139 cm at A15, 130–234 cm at A17), so a conductivity that continues to increase below ∼1.5 m — continued compaction rather than a lateral site difference — could in principle mimic the contrast. Two internal diagnostics limit this alternative. Within the A17 set itself, which spans 1.1 m, a depth-constant K_d leaves only the 0.40 K residual RMSE of Table 2 and predicts the withheld deepest sensor (234 cm) to 0.25 K (§3.3); a $K(z)$ still rising across that span would curve the profile detectably. The borestem-cut sweep similarly leaves both retrievals unchanged as the A15 set is restricted to its deepest five sensors ($z_b = 90$ cm, §3.2). Depth-dependent compaction below the probed range cannot be excluded with two boreholes, but it is not required by the data, whereas the compositional difference between the sites is independently established.

A17 remains the more conductive site at the published basal heat fluxes and across moderate revisions; only the most adverse corner of the independent-Q_b envelope (§4.2) erases the contrast.

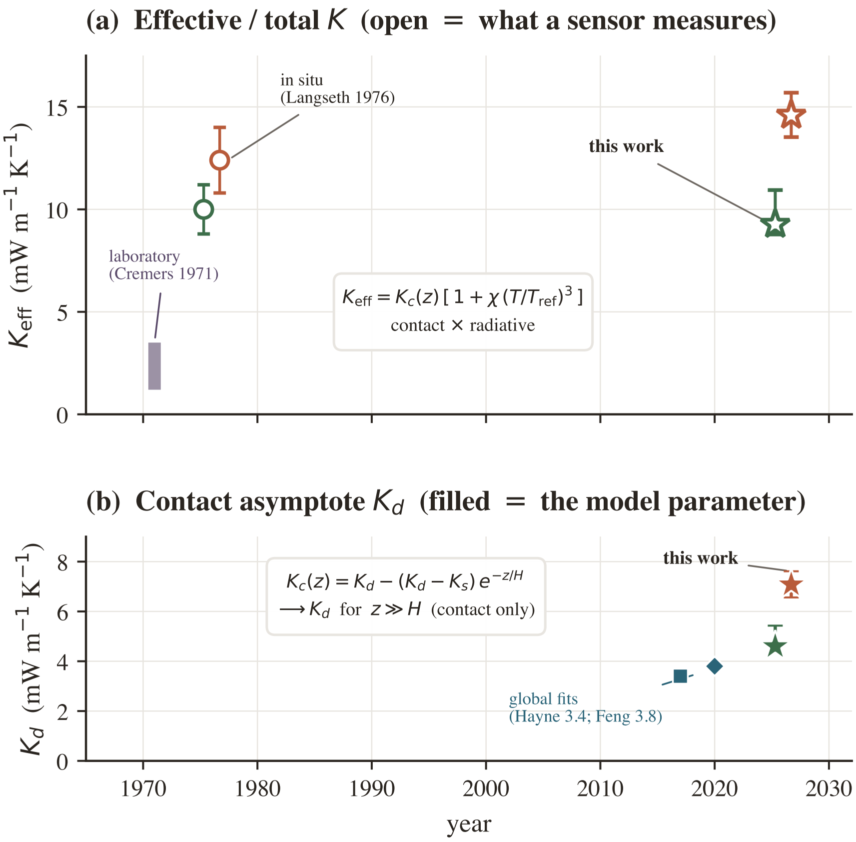

Figure 11. Five decades of meter-scale lunar conductivity estimates, split by definition class so the two are never conflated. **(a)** Effective / total K — what a sensor at depth measures, radiative term included (open markers): laboratory Apollo 12 fines (1.2–3.5 mW m^{-1} K^{-1}; Cremers and Birkebak, 1971) and the revised in-situ HFE reductions (≈10 at A15, ≈12 at A17; Langseth et al., 1976, their Fig. 6). **(b)** Contact asymptote K_d — the Hayne-form model parameter (filled markers): the global Diviner fits of Hayne et al. (2017) ($K_d = 3.4$) and Feng et al. (2020) ($K_d = 3.8$). This work appears in both panels: it is one retrieval expressed in each definition, with $K_{\rm eff} \approx 2\,K_d$ because the radiative term is carried in the effective K but not in the contact asymptote. Error bars are 1σ bootstrap. Every literature value was read from the cited paper.

Within every admissible combination of composition and porosity the sign is unchanged, even where the absolute A17 value is not.

4.5 Limitations and caveats

Six caveats accompany any use of the result. *(i)* With only two in-situ boreholes, the inter-site difference is insufficient to establish regional heterogeneity across the lunar surface; distinguishing genuine regional variability from local heterogeneity at two specific borehole locations requires additional landed thermometry. *(ii)* The meter-scale temperature gradient constrains the ratio Q_b/K_d, not K_d alone (§4.2); the data favor but do not establish a conductivity contrast over a uniform conductivity with differing basal flux. *(iii)* The contrast magnitude is not resolved at the 95% level even at the published basal heat fluxes ($\Delta K_d^* = 2.3$, bootstrap CI $[-0.1, 3.6]$ spanning zero, $P_{\rm boot}{\approx}0.03$; $\sim$97% of bootstrap draws favor the A17 ordering). Its *ordering*, however, is robust: on the directly solved RMSE(K_d, Q_b) surface the joint posterior keeps $P(K_d^{\rm A17} > K_d^{\rm A15}) \geq 99\%$ across the Saito et al. (2007); Nagihara et al. (2018) Q_b reanalysis envelope, and even the most adverse independent Q_b pairing only reduces the contrast to $\sim$+1.5 mW m^{-1} K^{-1} without erasing it. What the flux debate does forfeit is the *absolute* magnitude of both K_d values and of their contrast. These two robustness statements — the $\sim$97% sensor bootstrap and the $\geq$99% Q_b-envelope ordering — address orthogonal uncertainty axes and are not combined into a single joint probability; a hierarchical treatment carrying both simultaneously (caveat vii) would leave the sign intact but widen the interval. *(iv)* The retrieval adopts the Hayne et al. (2017) functional form for $K(T, z)$;

alternative piecewise parameterizations are not exhausted, and a full per-site recalibration of the Martínez and Siegler (2021) coefficients (whose density lever is already admissible at both sites but saturates at A17, §4.3) is the immediate next step. *(v)* The diurnal-amplitude depth profile (Fig. 4) serves only as a borestem diagnostic and not as a frequency-domain validation, because below 80 cm both models predict amplitudes well below the 1971–1977 digitization noise floor. *(vi)* The insolation model is the synodic cosine forcing of §2 with no local topography. This is least realistic at Apollo 17, whose probes sit on the Taurus–Littrow valley floor between massifs that shadow the site near the terminators and re-radiate toward it at midday (Langseth et al., 1976); terrain effects of this kind plausibly contribute to the +4.1 K surface-temperature bias in the Diviner comparison (§3.3). The retrieval fits only the sub-skin *lunation-mean* temperatures, so terrain enters through its lunation-mean perturbation; that perturbation is not negligible a priori — horizon blocking removes of order 1% of the lunation-mean insolation at a valley-floor site, the same order as the albedo envelope of Table 4 — but it is partially offset by terrain infrared emission and reflected sunlight, which return flux of the same order with the opposite sign. A terrain-resolved forcing carrying both channels is the natural refinement for any future re-analysis. *(vii)* The stability windows end at different epochs across sensors while the record is still warming at depth (Nagihara et al., 2018), so the per-sensor T_{eq} values mix epochs; the bootstrap resamples sensors as exchangeable, ignoring probe-level calibration offsets and common temporal drift. The common-epoch sensitivity is now quantified (§3.3): A15 is unaffected, while A17 moves by -0.89 mW m^{-1} K^{-1} (carried as σ_{epoch} in Table 4) and the contrast becomes more marginal ($P_{\mathrm{boot}} = 0.04$) while retaining its sign. A probe-clustered (hierarchical) uncertainty treatment remains follow-on work.

5 Conclusions

The two Apollo boreholes require distinctly different meter-scale conductivities. Holding the Hayne et al. (2017) functional form fixed and sweeping K_d alone against the meter-scale-sensor RMSE, the per-site retrieval yields $K^*_{d,\mathrm{A15}} = 4.60^{+0.86}_{-0.21}$ and $K^*_{d,\mathrm{A17}} = 7.08^{+0.53}_{-0.52}\,\mathrm{mW/(m\,K)}$ (1σ non-parametric bootstrap, $N_{\mathrm{boot}} = 1500$) at the published basal heat fluxes, with the inter-site contrast only marginally resolved ($\Delta K^*_d = 2.3$, 95% CI $[-0.1, 3.6]$ spanning zero, $P_{\mathrm{boot}} \approx 0.03$). The meter-scale-sensor RMSE decreases from 1.09 K and 0.89 K under the published global $K_d = 3.4\,\mathrm{mW/(m\,K)}$ to 1.00 K and 0.40 K under the per-site retrieval; the Martínez and Siegler (2021) $K(T, \rho)$ forward evaluation, calibrated against the same family of orbital products, reproduces both sites to within $\sim$1 K without per-site freedom, though it cannot match the sharpened per-site A17 fit (Table 2). The inter-site requirement is robust to the borestem exclusion depth, the stability-window threshold, a joint K_d–H fit, and the independent Diviner surface-temperature closure check.

The absolute K^*_d values are conditional on the adopted Q_b. Because the meter-scale temperature gradient constrains the ratio Q_b/K_d rather than K_d alone, the inter-site difference admits two interpretations: a conductivity contrast or a basal-flux contrast. A pooled three-model comparison cannot separate the two readings (ΔAICc ≤ 0.5 across all three models); the per-site evidence at A17 is what is decisive, ΔAICc ≈ 23 for the site fit over the global value (Table 3) without excluding the alternative. The quantity constrained tightly at each borehole is the meter-scale temperature gradient itself.

The immediate application is to forward thermal models used by upcoming lunar sub-surface RTM retrievals. These retrievals require a meter-scale $T(z)$ boundary condition that orbital-only conductivity models cannot supply on a per-site basis. The per-site K^*_d values reported here, evaluated in the Hayne thermal column, provide that boundary condition (§1); the upcoming TSUKIMI terahertz mission (Kasai et al., 2022) is one immediate beneficiary. A full per-site recalibration of the Martínez and Siegler (2021) form, and the extension of the meter-scale $T(z)$ column to include surface-shadowing effects on the RTM side, are both deferred to follow-on work. Extending the analysis beyond two boreholes, to a regionally resolved treatment of the meter-scale lunar thermal state, is fundamentally a measurement problem that requires meter-scale temperature data from additional landed instruments (e.g. Chang'E and ChaSTE-class experiments).

Acknowledgments

The authors thank the original Apollo HFE Principal Investigators and the Nagihara et al. (2018) team for restoring the 1975–1977 portion of the 1971–1977 record from the original tapes (the 1971–1975 portion was retained from the original PI processing), without which this work would not have been possible.

Open Research

Availability Statement. The restored Apollo 15 and 17 Heat-Flow Experiment temperature record (1971–1977) analyzed here is openly available as the supporting information of Nagihara et al. (2018) (`https://doi.org/10.1029/2018JE005579`). The Diviner Lunar Radiometer Global Cumulative Product used for the out-of-sample surface-temperature closure (Williams et al., 2017) is available from the NASA Planetary Data System Geosciences Node (`https://pds-geosciences.wustl.edu`; LRO-L-DLRE-5-GCP-V1.0, `https://doi.org/10.17189/1520650`). The repository bundles the NASA SPICE Toolkit kernels (JPL DE440; Acton, 1996; Acton et al., 2018) used by its ephemeris module. The 1-D Crank–Nicolson heat solver, the ephemeris-driven solar-forcing routines, the K_d retrieval and bootstrap code, the validation diagnostics, and the data and scripts needed to reproduce every figure and table in this manuscript are archived at `https://github.com/rp3gregorio/Lunar-HFE`; a Zenodo-archived release with a citable DOI will be issued at the time of acceptance and added to the reference list.

Conflict of Interest Disclosure

The authors declare there are no conflicts of interest for this manuscript.

Nomenclature

Symbols.

Symbol	Definition	Value / units	Source
K_d	deep asymptote of the contact K_c (retrieved); effective K at sensors $\approx 2\times$	4.60, 7.08 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
K_s	surface regolith thermal conductivity	7.4×10^{-4} $\mathrm{W\,m^{-1}\,K^{-1}}$	Hayne et al. (2017)
H	e-folding depth of $K(z)$ transition	0.06 m	Hayne et al. (2017)
χ	radiative conductivity multiplier coefficient	2.7	Hayne et al. (2017)
T_{ref}	normalization temperature for radiative term	350 K	Hayne et al. (2017)
ρ_s, ρ_d	surface / asymptotic regolith bulk density	1100, 1800 $\mathrm{kg\,m^{-3}}$	Hayne et al. (2017)
$c_p(T)$	specific heat (4th-order polynomial in T)	$\mathrm{J\,kg^{-1}\,K^{-1}}$	Hemingway et al. (1973); Hayne et al. (2017)
Q_b	basal heat flux	21 (A15), 16 (A17) $\mathrm{mW\,m^{-2}}$	Langseth et al. (1976); Nagihara et al. (2018)
S_0	solar constant (irradiance at 1 AU)	1361 $\mathrm{W\,m^{-2}}$	Kopp and Lean (2011)
A, ε	broadband Bond albedo, emissivity	site-specific; 0.95	Vasavada et al. (2012); Bandfield et al. (2015)
σ	Stefan–Boltzmann constant	5.6704×10^{-8} $\mathrm{W\,m^{-2}\,K^{-4}}$	CODATA 2018
$\theta_\odot(t)$	solar zenith angle at site	rad	synodic approximation, §2
$F_\odot(t)$	incident solar irradiance at surface	$\mathrm{W\,m^{-2}}$	this work, eq. 1
$T(z,t)$, T_s	subsurface and skin ($z=0$) temperature	K	this work
z	depth coordinate (positive downward)	m	—
80 cm	borestem-exclusion depth (*a priori* cut)	80 cm	this work, §2.4
δ	diurnal thermal skin depth	∼3–5 cm	Hayne et al. (2017)
ΔK_d	inter-site contrast $K_d^{\mathrm{A17}} - K_d^{\mathrm{A15}}$	2.31 $\mathrm{mW\,m^{-1}\,K^{-1}}$	this work
RMSE	meter-scale-sensor root-mean-square residual	K	this work
N	number of meter-scale sensors	7 (A15), 16 (A17)	Nagihara et al. (2018)
N_{boot}	non-parametric bootstrap resamples	1500	Efron and Tibshirani (1993)
P_{boot}	bootstrap tail proportion $P(\Delta K_d^* \le 0)$ (not a null-centered p-value)	0.031	this work

Subscripts.

s	surface (regolith, skin)
d	asymptotic (deep) regolith
b	basal (lower BC)
ref	reference (normalization)
$\odot$	solar
eq	stability-window mean
boot	bootstrap quantity

Superscripts.

*	best-fit / retrieved (e.g. K_d^*)
$^{\mathrm{A15,A17}}$	per-site qualifier
$^{\mathrm{ref}}$	reference value
$^{\mathrm{HFE}}$	Apollo HFE-measured
$\langle\cdot\rangle$	stability-window time average

Acronyms.

HFE	Heat-Flow Experiment (Apollo 15 and 17 buried thermometers)
TG / TR	gradient-bridge / ring-bridge sensor (HFE sensor types)
LST	local solar time
Diviner GCP	Diviner Lunar Radiometer Global Cumulative Product
MCMC	Markov-chain Monte Carlo
BCa	bias-corrected accelerated bootstrap
SPICE	NASA ancillary information system for planetary geometry
DE440	JPL planetary and lunar ephemeris release 440
RTM	radiative-transfer model
TSUKIMI	lunar mission consuming the meter-scale $T(z)$ retrieved here

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